

OCR Error Recovery in Egyptian Medicine Search

What was tested, what failed, and why

medicine-search-clean engineering analysis

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Abstract

This report tests whether Algorithm 4, the medicine-family search system, can recover an Egyptian medicine family from text already produced by an OCR model. The input is an OCR string, not a prescription image. Across all 595 observations, Algorithm 4 reaches 50.25% Hit@1 (verified family ranked first) and 73.78% Hit@20 (verified family within the first 20). The primary denominator, which is the score to cite, removes 17 queries that exactly name a different real drug and gives each distinct query-target pair one vote; its 464 cases reach 47.63% Hit@1 and 71.12% Hit@20. An extreme prediction has compact Levenshtein distance, calculated after removing spaces and punctuation, divided by target length above 0.60. All 121 extreme predictions remain in the inclusive 595-row diagnostic view. Compact distance four or more creates 140 of its 156 top-20 misses.

1 What Was Tested

The experiment starts after OCR. Fourteen OCR models had already read medicine images and written their text predictions to `predictions.csv`. Algorithm 4 receives one prediction at a time. The verified family is hidden until scoring.

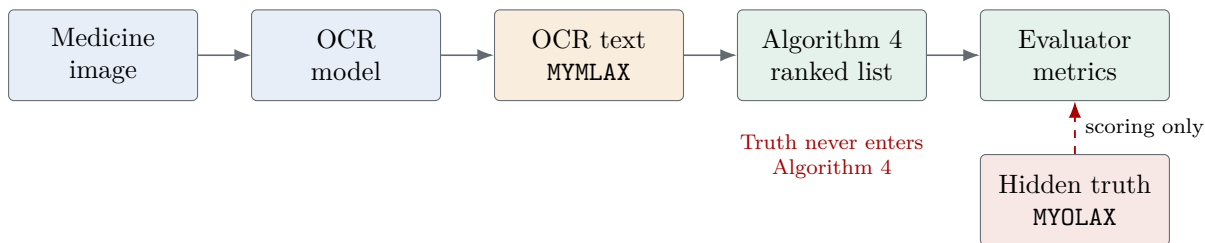


Figure 1: Read left to right: the OCR output MYMLAX enters Algorithm 4, while hidden truth MYOLAX reaches only the evaluator. The benchmark therefore measures medicine-search recovery after OCR; it does not rerun OCR or evaluate prescription segmentation.

Insights

1. The benchmark isolates search recovery from OCR recognition quality.
2. Keeping the verified target outside Algorithm 4 prevents answer leakage during retrieval.

The 595 rows are OCR *observations*, not necessarily 595 unique images. Several models can produce predictions for the same medicine. The report has 477 unique normalized query-target pairs and 118 repeated observations.

Stage	Concrete example	Meaning
Input	MYMLAX, model = EasyOCR	Text delivered to Algorithm 4.
Process	Search 25,066 catalog medicines	Generate and rank family candidates.
Output	MYOLAX at rank 1	The verified family is recovered first.
Scoring	H@1 = 1, H@20 = 1	The evaluator compares output with hidden truth.

Table 1: The example traces one row from OCR input MYMLAX through a 25,066-medicine search to MYOLAX at rank 1. The expected family affects metrics, never retrieval.

These repeated rows change which denominator answers which question. The next section fixes the counting rules before any aggregate score is interpreted.

2 Data Contract and Definitions

An **observation** is one OCR model output paired with one verified medicine target. For example, EasyOCR output MYMLAX paired with verified target MYOLAX is one observation. Another model producing the same pair is another observation, but not another unique query-target pair.

Dataset object	Size	Meaning
Source prediction CSV	595 rows $\times$ 15 columns	OCR text, verified target, OCR model, lengths, source alignment counts, normalized distances, and operation sequence. No source cell is missing.
Algorithm result CSV	595 rows $\times$ 44 columns	One joined search result per source row, including candidates, ranks, Hit@k, cohort, split, decision, and latency.
OCR source models	14	Models contribute 26–73 rows each; the mean is 42.5 rows per model.
Verified target families	27	Targets contribute 1–92 rows each; the mean is 22.04 rows per target.
Unique raw OCR strings	481	Repeated strings remain separate observations. Mean repetition is 1.24 rows per raw string.
Unique compact query-target pairs	477	Spaces, punctuation, and case are removed before duplicate pairs are identified.
Scored observations	578	The 17 real-drug collisions remain diagnostic rows but do not enter the primary correctness denominator.
Primary scored unique pairs	464	Each scored compact query-target pair receives one vote. This is the primary benchmark denominator.

Table 2: The two CSV files join one-to-one. “Row count”, “unique-pair count”, and “scored count” are different denominators and therefore produce different accuracy values.

Which number should you report? Cite the primary 464-pair result, 47.63% Hit@1 and 71.12% Hit@20, when comparing search algorithms because each distinct scored query-target pair receives one vote. Use the inclusive 595-row result, 50.25% Hit@1 and 73.78% Hit@20, when auditing every supplied OCR observation, including repetitions and real-drug collisions.

The **compact form** lowercases text and removes spaces, punctuation, and other non-alphanumeric characters. Thus CLEX A NE and CLEXANE share the compact form **cllexane**. Let q_c be the compact OCR query and y_c the compact verified target. The search-side distance and normalized distance are

$$d = L(q_c, y_c), \quad r = \frac{d}{\max(|y_c|, 1)},$$

where L is Levenshtein distance. One insertion, deletion, or replacement costs one. A row is **extreme** when $r > 0.60$, unless the query exactly names a different catalog family, which takes priority as a real-drug collision. For GHONT $\rightarrow$ CALAMINE, $d = 7$, $|y_c| = 8$, and $r = 7/8 = 0.875$; the row is extreme.

The 0.40 and 0.60 boundaries are fixed operational review bands inherited from the benchmark methodology. They were not fitted to Algorithm 4 results and do not represent clinical risk probabilities. The original filtering policy sent $r > 0.60$ rows to manual review. This analysis keeps those rows and reports their performance instead of rejecting them.

The source CSV also supplies a character alignment that can retain spaces. Its `edit_distance_count` can therefore differ slightly from compact distance d . Figures explicitly say *source distance* or *compact distance*; the terms are not interchangeable.

Representation	OCR text	Target	Calculation	Result
Source alignment	CLEX A NE	CLEXANE	two retained spaces	$d_s = 2, 2/7 = 0.286$
Compact search	cllexane	cllexane	spaces removed first	$d = 0, r = 0$

Table 3: One OCR pair can have nonzero source distance but zero compact distance. Source plots describe the supplied alignment; search cohorts use compact d and r .

Term	Definition used in this report
n	Number of rows in the displayed group. It is the denominator for a group percentage.
Mean	Sum of values divided by n . Large outliers can move it.
Median	Middle value after sorting. Half the rows are below it and half are above it.
Q1-Q3	Middle 50% of rows, from the 25th to the 75th percentile.
Hit@k	Percentage of rows whose verified family appears at rank k or earlier.
Failure rate	Top-20 misses in a group divided by rows in that group.
Failure share	Top-20 misses in a group divided by all top-20 misses.

Table 4: Every average, rate, and percentage in later figures uses one of these definitions. Group plots print n beside the group name.

Target-family hashing keeps KETOROLAC’s 92 rows in development for analysis and tuning, and VIGOREX’s 32 rows in holdout for unseen-family checking. Other label examples are SAFE/one-edit KETONOLAC $\rightarrow$ KETOROLAC, two-edit ABISAGLOR $\rightarrow$ ABASAGLAR, multi-edit ABASCOLOGY $\rightarrow$ ABASAGLAR, visible-fragment RIVOTRI $\rightarrow$ RIVOTRIL, CAUTION/high-distance OSMOL $\rightarrow$ OSTOCAL, and DANGEROUS/collision RIVOTRIL $\rightarrow$ RIVO. An ambiguous ABISAGLOR $\rightarrow$ ABASAGLAR response lists candidates; no-match H $\rightarrow$ RIVO returns zero. Decision examples are possible KETONOLAC, equal-distance OSTEOCL $\rightarrow$ OSTOCAL, and family-variant ABISAGLOR.

OCR benchmark at a glance

All panels describe the same 595 joined OCR observations before collision rows are removed from primary scoring.

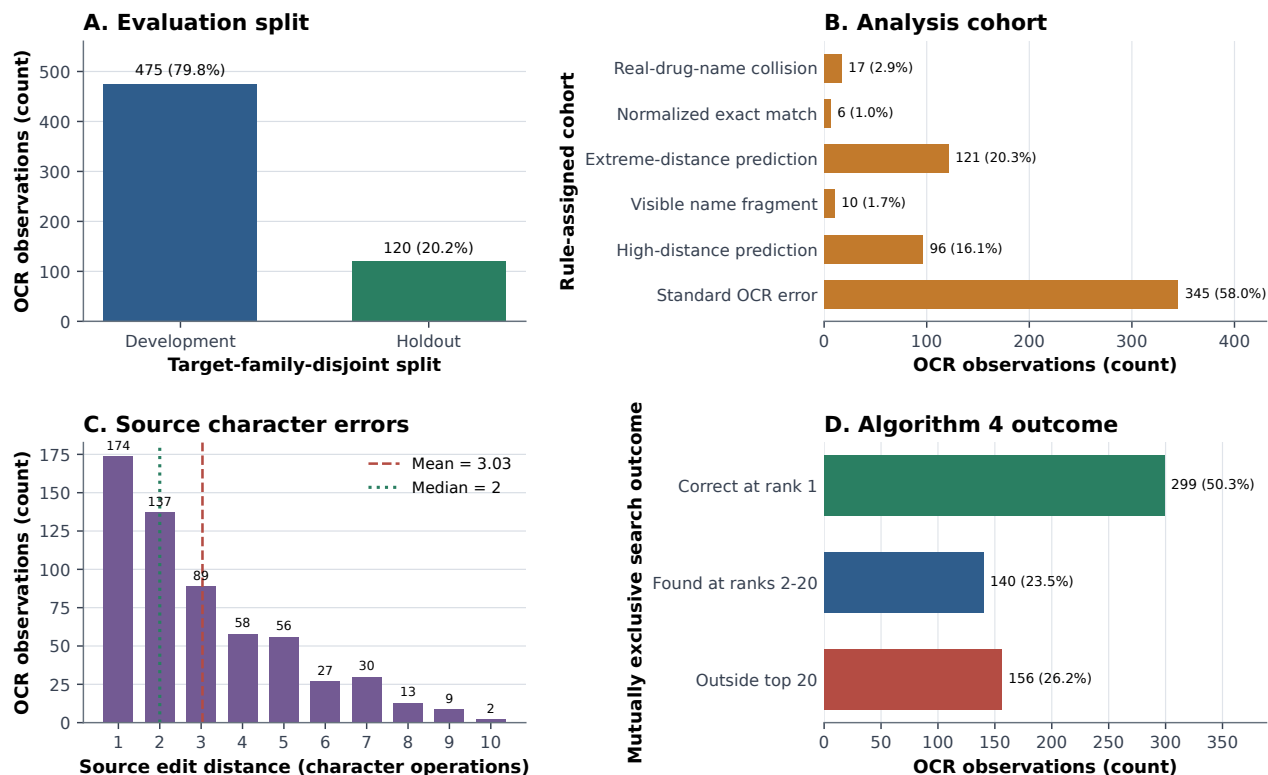


Figure 2: Panels A and B count observations by target-family split and cohort; panel C counts observations at each source edit distance; panel D partitions Algorithm 4 outcomes. The CSV is dominated by standard OCR errors, but extreme predictions form one fifth of all rows and most are outside the top 20. Cohort examples are standard ABISAGLOR→ABASAGLAR, high-distance OSMOL→OSTOCAL, extreme GHONT→CALAMINE, fragment RIVOTRI→RIVOTRIL, collision RIVOTRIL→RIVO, and compact exact TO CO→TOCO. In panel C, mean 3.03 exceeds median 2 because a small severe tail pulls the mean upward.

Insights

1. The 20.2% family-disjoint holdout provides a genuine unseen-family check rather than a random-row split.
2. High and extreme corruption together cover 36.4% of the CSV, so mild typo performance cannot dominate the benchmark.
3. Mean source distance exceeds the median because a small severe tail changes the aggregate.
4. The 140 ranking failures and 156 retrieval failures require different algorithm changes.

Numeric field	Mean	Median	Q1	Q3	Min–max
OCR-output length (characters)	7.24	7.00	6.00	9.00	1–13
Verified-target length (characters)	7.27	8.00	6.00	9.00	4–9
Source edit distance (operations)	3.03	2.00	1.00	4.00	1–10
Source distance / target length	0.434	0.333	0.200	0.556	0.111–2.000
Compact edit distance (operations)	2.96	2.00	1.00	4.00	0–10
Compact distance / target length	0.425	0.333	0.200	0.556	0.000–2.000
Source similarity	0.566	0.667	0.444	0.800	–1.000–0.889
Returned candidate count	28.66	31.00	25.00	34.00	0–44
Query latency (ms)	15.30	14.27	10.45	18.88	0.04–39.83

Table 5: Global descriptive statistics over all 595 observations. A normalized distance can exceed 1 when the number of edits exceeds the target length; similarity then becomes negative.

Categorical field	Complete distribution	Assignment rule
Split	Development 475; holdout 120	Deterministic hash of target family; one target cannot cross splits.
Danger	SAFE 342; CAUTION 236; DANGEROUS 17	DANGEROUS for real-drug collision; CAUTION for compact query length ≤ 4 or $r > 0.40$; SAFE otherwise.
Mistake type	1 edit 172; 2–3 edits 217; multi-edit 172; real-drug collision 17; prefix/suffix fragment 17	Collision and fragment rules first; otherwise compact edit count.
Response status	Ambiguous 583; no match 12	Ambiguous returns candidates without certifying one; no match returns zero candidates.
Decision type	Possible matches 330; equal-distance ambiguity 155; family-variant selection 98; no match 12	Algorithm 4 response branch after candidate generation and family grouping.
Search outcome	Rank 1 success 299; ranks 2–20 success 140; outside top 20 156	Position of the verified family in the returned family ranking.

Table 6: Complete distributions for the major categorical columns used in the paper. Counts in each row sum to 595. Cohort counts appear separately in Table 8. Mistake type describes the compact edit pattern; cohort applies severity and collision priority, so their fragment counts need not match.

The contract now fixes every denominator and label; the next section follows one OCR row through retrieval and scoring so those definitions become concrete.

3 One Complete Example

EasyOCR produced MYMLAX; the verified family is MYOLAX. Five characters match and one character is substituted. Algorithm 4 returns 36 candidates and places MYOLAX first.

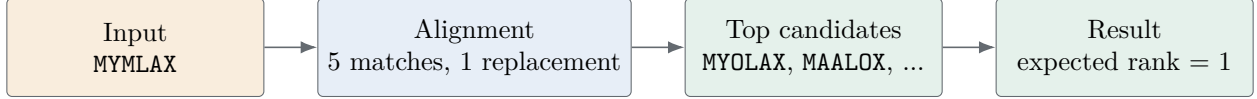


Figure 3: Read left to right: one replacement changes OCR input MYMLAX to verified family MYOLAX, which Algorithm 4 ranks first among 36 candidates. This is a successful one-edit recovery.

Insights

1. A single internal replacement preserves enough evidence for exact family recovery at rank 1.
2. Candidate generation is broad, 36 candidates, but the evaluation rewards the verified family’s final rank.

OCR input	Expected	Top 1	Rank	Failure type
MYMLAX	MYOLAX	MYOLAX	1	Full success.
KELOLAC	KETOROLAC	KETOLAC	2	Retrieval succeeds; ranking fails.
RIV	RIVOTRIL	RIVO	10	Fragment keeps weak but useful evidence.
GHONT	CALAMINE	GEOCAND	> 20	Severe OCR corruption removes target evidence.
H	RIVO	none	> 20	One character cannot identify a family.
RIVOTRIL	RIVO	RIVOTRIL	> 20	OCR output is another exact real drug name.

Table 7: The benchmark contains ranking errors, retrieval errors, insufficient evidence, and real-name collisions. They need different fixes.

For query q , $\text{Hit}@k$ equals one when the verified family occurs within the first k returned families:

$$\text{Hit}@k(q) = \mathbf{1}[R_k(q) \cap y(q) \neq \emptyset].$$

H@1 failure with H@20 success is a ranking failure. H@20 failure is a retrieval failure.

Every statistical figure is generated directly from the two benchmark CSV files. Each plot labels its x-axis, y-axis, unit, legend, and subgroup denominator inside the figure. Captions state the conclusion rather than redefining the axes.

One row explains the scoring mechanics; the next section checks how repeated strings, unequal model coverage, and unequal target coverage weight the full CSV.

4 Dataset Composition

Unless a figure says *primary*, this section uses all 595 observations: 475 development rows and 120 holdout rows. The split hashes the verified target family, so all observations for one family stay in one split. This prevents the same family from appearing in both development and holdout data. For example, all

92 KETOROLAC observations are in development, while all 32 VIGOREX observations are in holdout. Development supports analysis and tuning; holdout checks whether conclusions survive on families excluded from that work.

Dataset inclusion and evaluation flow

Rows: 595 input, 595 mapped, 595 evaluated. Extreme means compact normalized distance > 0.60 .

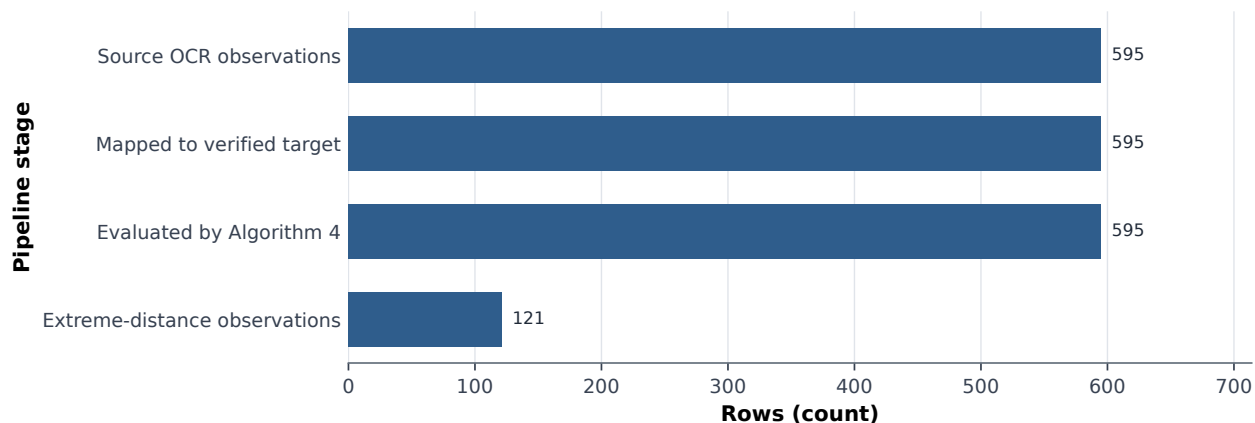


Figure 4: Horizontal bars count rows at each pipeline stage. Input, mapped, and evaluated counts all equal 595, so no OCR observation is removed; the separate 121-row extreme bar means compact distance divided by target length exceeds 0.60, not that those rows were rejected.

Insights

1. Equal input, mapped, and evaluated counts rule out hidden row filtering as a source of the reported score.
2. Extreme rows remain a measured failure cohort rather than being discarded as invalid predictions.

Cohort	Rule, applied in this priority order	<i>n</i>	Share
Real-drug collision	Compact OCR query exactly names a different Egyptian catalog family.	17	2.9%
Normalized exact	Compact query equals compact verified target.	6	1.0%
Extreme distance	No collision; compact normalized distance $r > 0.60$.	121	20.3%
Limited evidence	No earlier rule; fewer than two query characters, no shared character, or no shared bigram with fewer than two shared characters. No row meets this rule in this CSV.	0	0.0%
Visible fragment	No earlier rule; query has at least three characters and is a contiguous part of the target.	10	1.7%
High distance	No earlier rule; $0.40 < r \leq 0.60$.	96	16.1%
Standard OCR error	No earlier rule; $0 < r \leq 0.40$.	345	58.0%

Table 8: Cohorts are deterministic labels, not subjective judgments. The rules partition all 595 observations and sum to 100%.

Most frequently repeated OCR outputs

There are 481 unique raw strings; mean repetition is 1.24 rows per string.

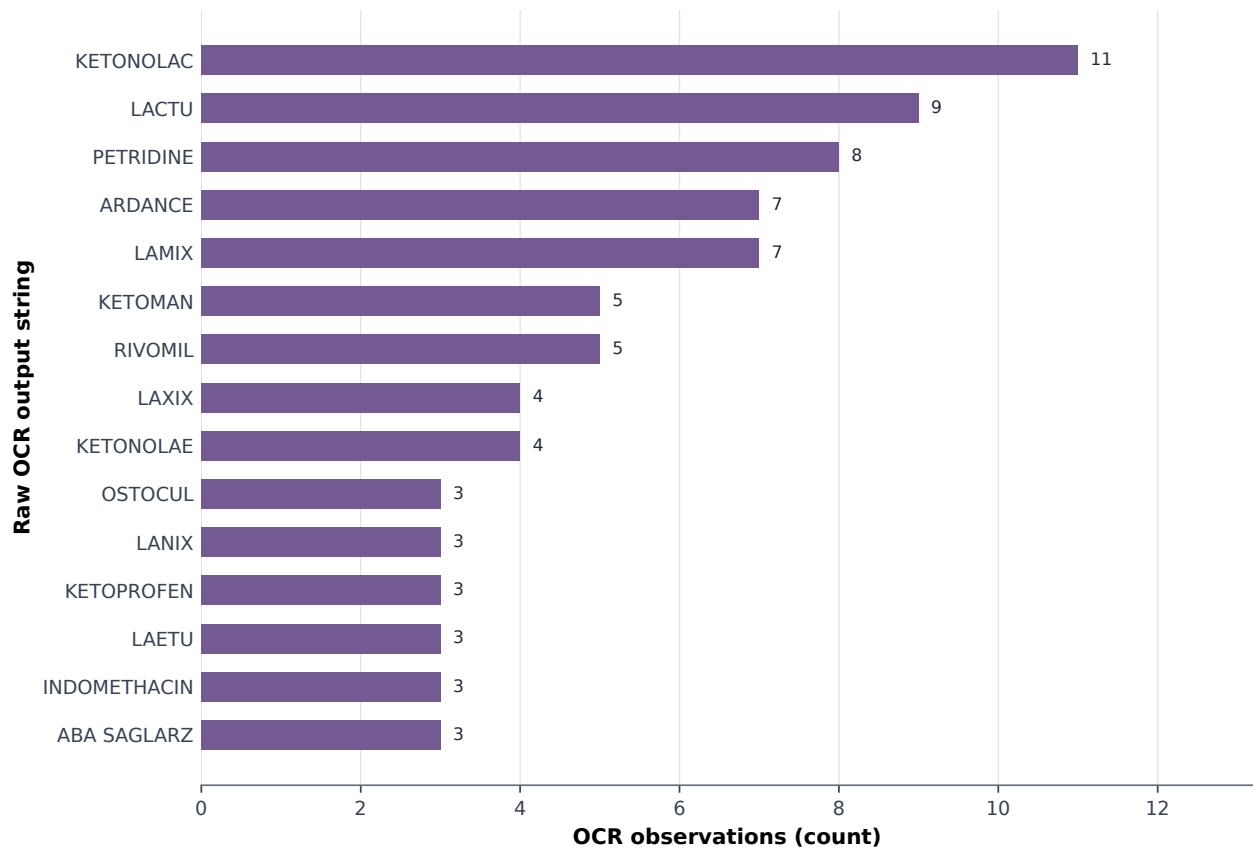


Figure 5: The x-axis counts repeated observations for each raw OCR string on the y-axis. For example, KETONOLAC occurs 11 times, so it receives 11 votes in the inclusive score but one vote after unique-pair deduplication.

Insights

1. Inclusive metrics weight frequently repeated OCR strings more heavily than rare strings.
2. Unique-pair scoring removes that exposure imbalance when comparing search algorithms.

Observation coverage by OCR source model

All 14 models are retained; mean coverage is 42.5 rows per model.

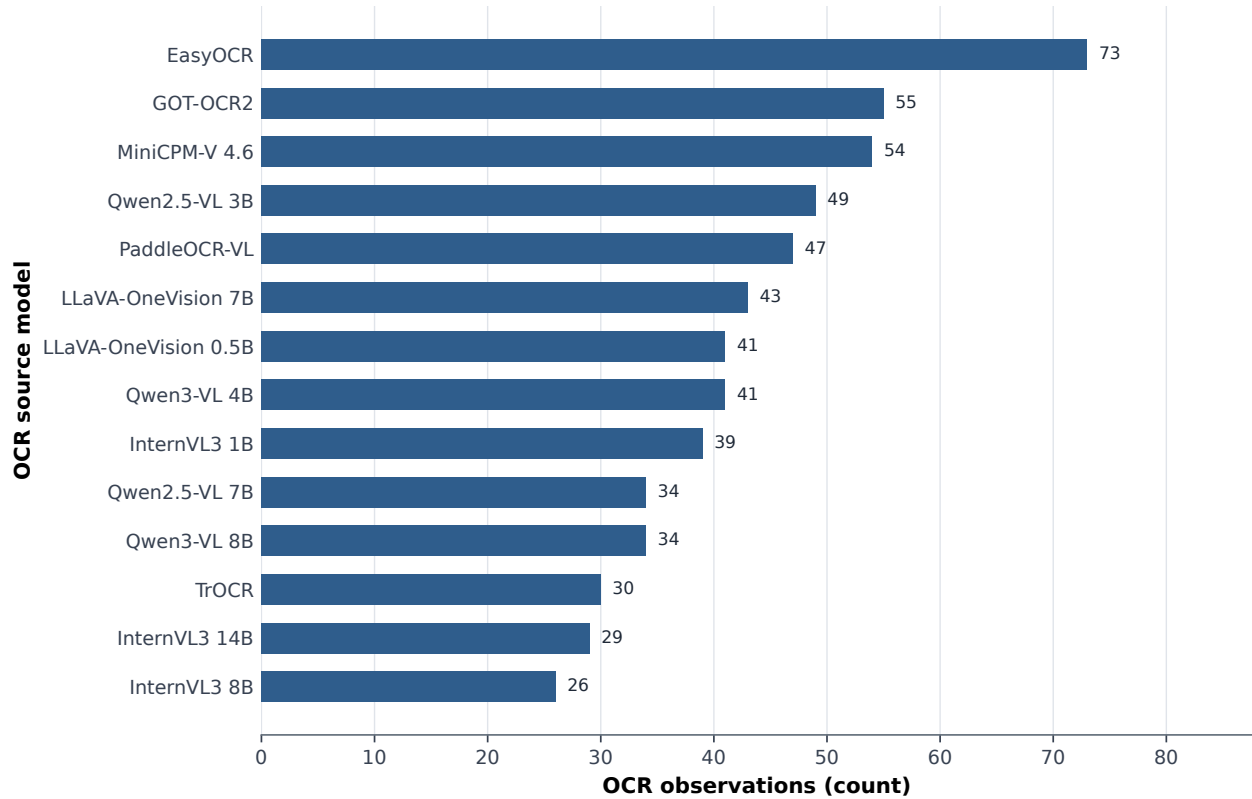


Figure 6: Each bar counts observations supplied by one OCR model. EasyOCR contributes 73 rows and InternVL3 8B contributes 26, so their downstream percentages use different denominators and are not a matched-image comparison.

Insights

1. Model-level search rates use unequal sample sizes and therefore have unequal statistical stability.
2. The coverage imbalance prevents interpreting later raw rates as a matched-image OCR leaderboard.

Observation coverage by verified medicine target

All 27 targets are shown; mean coverage is 22.0 rows per target.

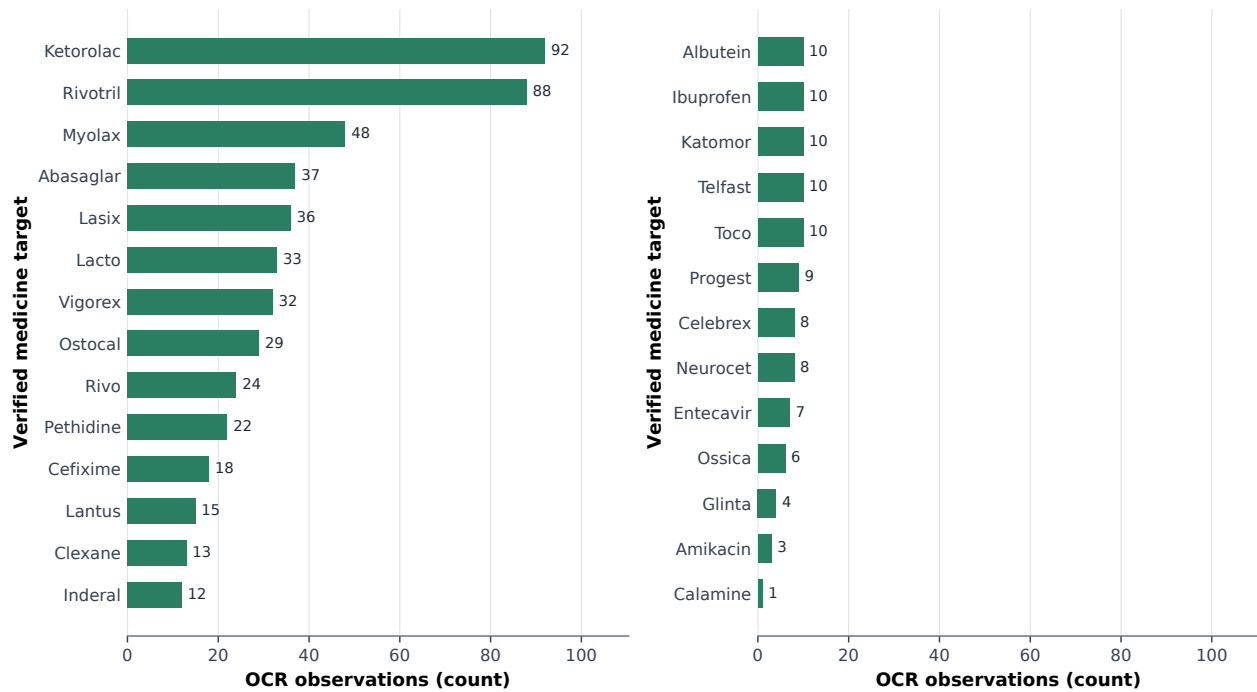


Figure 7: Each bar counts observations whose verified family is the named target. KETOROLAC has 92 rows and CALAMINE has 1; the aggregate score is therefore much more sensitive to KETOROLAC and RIVOTRIL than to low-count targets.

Insights

1. High-volume targets dominate the inclusive score and the absolute number of failures.
2. Percentages for targets with one or a few rows are diagnostic observations, not stable estimates.

Observation count by analysis cohort

Priority: collision, exact, >0.60 extreme, limited evidence (n=0), fragment, >0.40 high, otherwise standard.

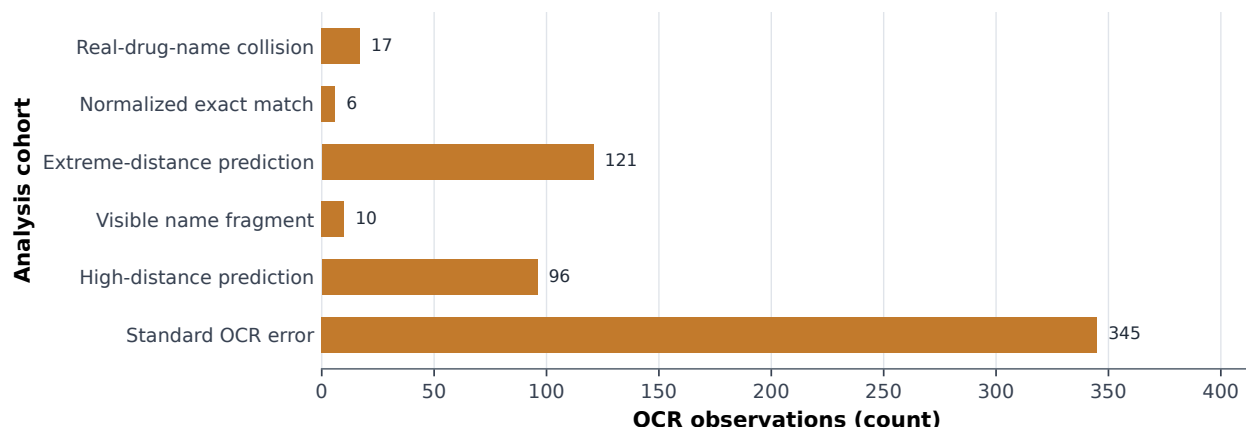


Figure 8: Bars count mutually exclusive cohort assignments after collision and exact-match priority: standard has $r \leq 0.40$, high has $0.40 < r \leq 0.60$, and extreme has $r > 0.60$; exact means compact strings equal, fragment means the query is a target substring, and collision means the query exactly names another family. Standard contributes 345 rows, but the 121 extreme rows are large enough to depress the inclusive score.

Insights

1. Standard errors are the majority, but high and extreme rows are too common to treat as edge cases.
2. Priority-ordered, mutually exclusive rules prevent one row from inflating several cohort counts.

The composition audit shows unequal coverage and a 121-row extreme cohort; the next section identifies the character operations and lengths behind that mix.

5 What the OCR Errors Look Like

The source CSV aligns each OCR string with its verified target before search. Let d_s be its supplied edit count, ℓ_q the supplied OCR-output length, and ℓ_y the supplied target length. The source fields plotted below are

$$d_s, \quad \frac{d_s}{\max(\ell_y, 1)}, \quad \frac{d_s}{\max(\ell_q, 1)}, \quad 1 - \frac{d_s}{\max(\ell_y, 1)}, \quad \ell_y - \ell_q.$$

These fields may retain spaces, while the compact search ratio r removes them. Every mean and median in this section is calculated across all 595 observations unless a figure states another denominator.

The alignment trace uses $M(c)$ for a matched character, $I(c)$ for a target character missing from the OCR output, $D(c)$ for an extra OCR character, and $R(a \rightarrow b)$ for a replacement needed to recover the target. The CSV columns `additions_count`, `deletions_count`, `flip_count`, and `matches_count` count these four operations.

Source edit-distance distribution

Insertions + deletions + replacements; mean=3.03, median=2 edits.

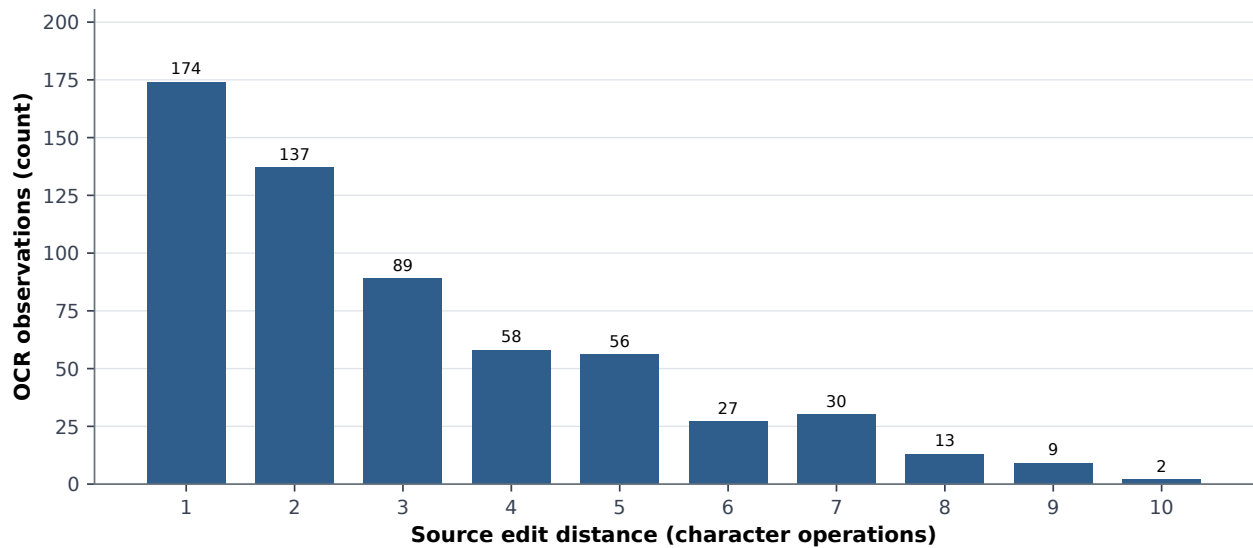


Figure 9: The x-axis is source edit count, including source spacing; bar height is the number of OCR observations. Distance 1 contains 174 rows, while only 2 rows reach distance 10; this severe tail raises the mean to 3.03 above median 2.

Insights

1. Median distance 2 describes the typical row better than mean 3.03 because the distribution is right-skewed.
2. The nonempty 7–10 edit tail identifies cases where ordinary typo retrieval is unlikely to work.

Edit distance normalized by verified target length

Source edits / target length; mean=0.434, median=0.333.

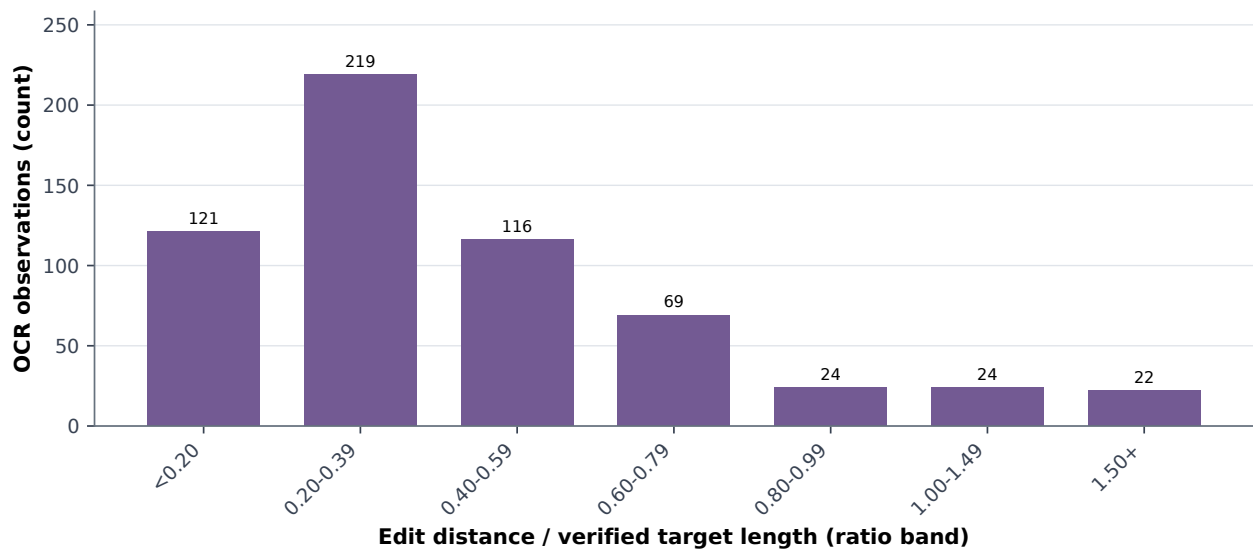


Figure 10: Bars count rows in bands of source edit count divided by source target length. Mean 0.434 and median 0.333 describe the source alignment; cohort labels instead use compact ratio r , which first removes spaces and punctuation.

Insights

1. Dividing by target length makes corruption severity comparable across short and long medicine names.
2. This source-alignment ratio should not be substituted for compact r because source spacing can change its value.

Edit distance normalized by OCR-output length

Source edits / OCR length; mean=0.482, median=0.333.

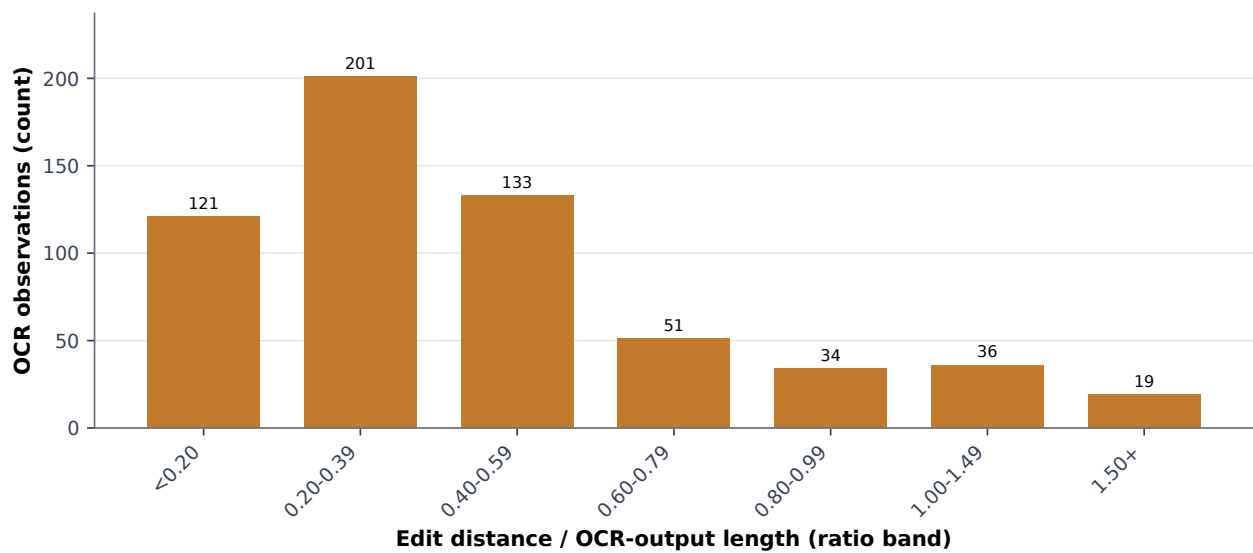


Figure 11: Bars count rows in bands of source edit count divided by OCR-output length. The smaller denominator on short outputs raises the mean to 0.482; the extreme value 9.0 remains visible in the 1.50+ band rather than being clipped.

Insights

1. Short OCR outputs create very large ratios because the observed text supplies the denominator.
2. The ratio is useful for diagnosing evidence loss but is too sensitive to output length to define the search cohorts.

Similarity to the verified target

Similarity = $1 - \text{source edits} / \text{target length}$; mean=0.566, median=0.667.

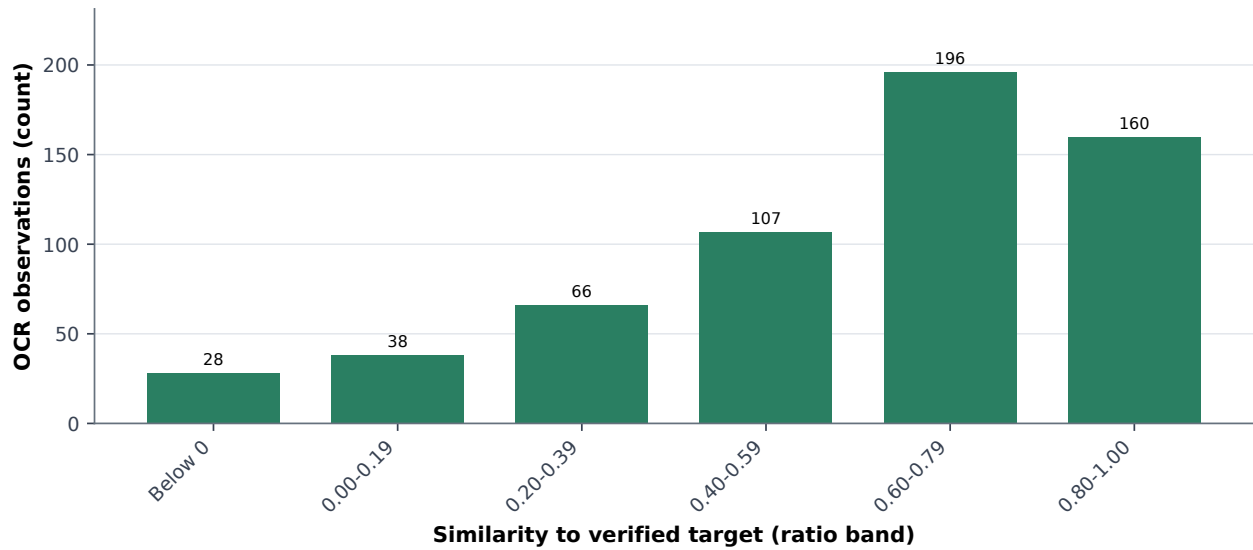


Figure 12: The x-axis bands source similarity $1 - d_s / \max(\ell_y, 1)$; bar height counts observations. Mean similarity is 0.566, median is 0.667, and negative values remain visible when edits exceed target length.

Insights

1. Similarity is a linear transform of normalized source distance, so it is a second view of the same signal rather than independent evidence.
2. Negative similarity values expose rows whose edit count exceeds the complete target length.

OCR-output length distribution

n=595; mean=7.24, median=7.0, range=1 to 13.

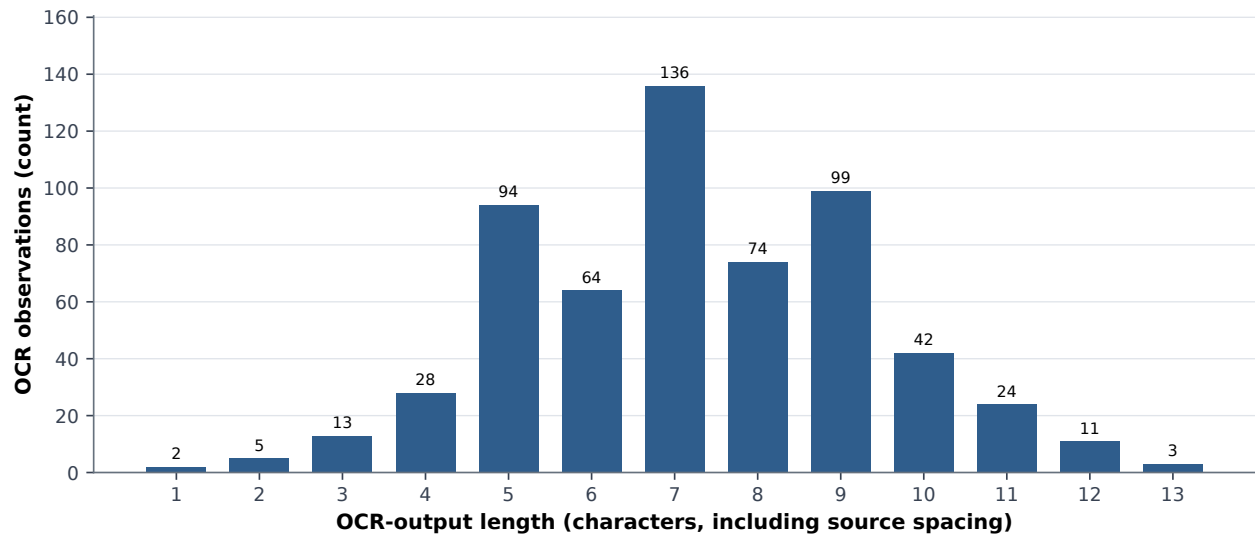


Figure 13: Bars count OCR observations at each source output length, including source spacing. Mean length is 7.24 and median is 7; the 20 rows with only 1–3 characters later reach just 10.00% Hit@1 and 15.00% Hit@20.

Insights

1. The 1–3 character tail is small but supplies too little lexical evidence for reliable family identification.
2. Length should trigger clarification behavior, not an automatic medicine choice.

Verified target length distribution

n=595; mean=7.27, median=8.0, range=4 to 9.

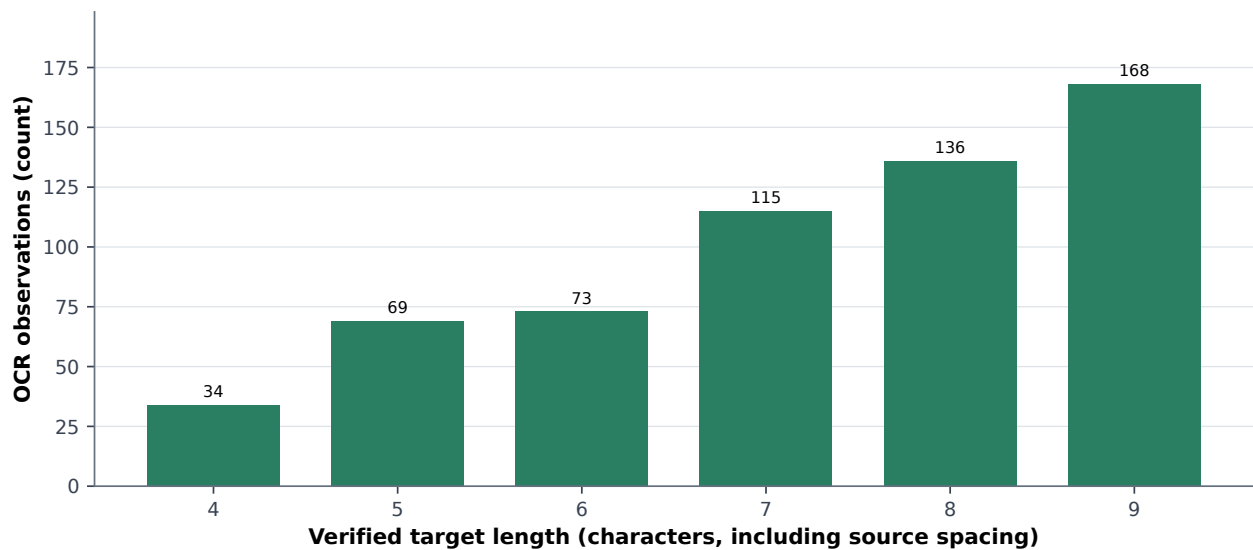


Figure 14: Bars count verified targets by source character length. Targets occupy only lengths 4–9, with mean 7.27 and median 8, while OCR outputs span 1–13; most length variance therefore comes from OCR corruption.

Insights

1. Verified names occupy a narrow 4–9 character range, while OCR output length varies much more widely.
2. Most extreme length variation is therefore produced by OCR corruption rather than target-name diversity.

Target-minus-output length difference

n=595; mean=0.03, median=0.0, range=-7 to 8.

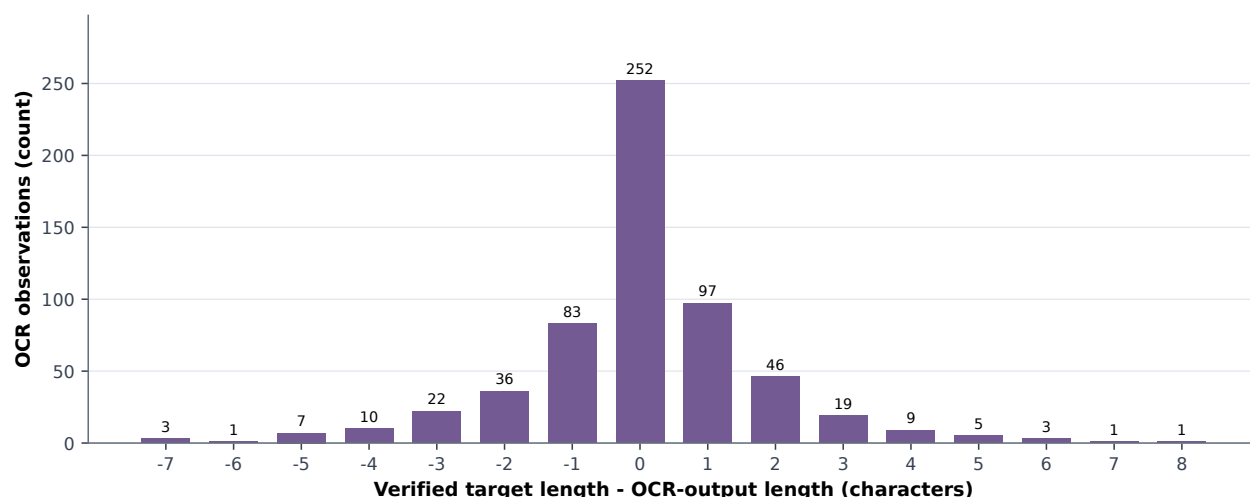


Figure 15: The x-axis is target length minus OCR-output length; positive means OCR is shorter and negative means it is longer. Mean 0.029 and median 0 show balanced length overall, but equal length can still contain replacements.

Insights

1. A near-zero mean length difference hides substantial shorter and longer tails that cancel each other.
2. Equal length cannot certify correctness because replacements preserve string length.

Character-alignment operation totals

Totals across 595 alignments; mean per row: additions 0.59, deletions 0.56, replacements 1.88, matches 4.80.

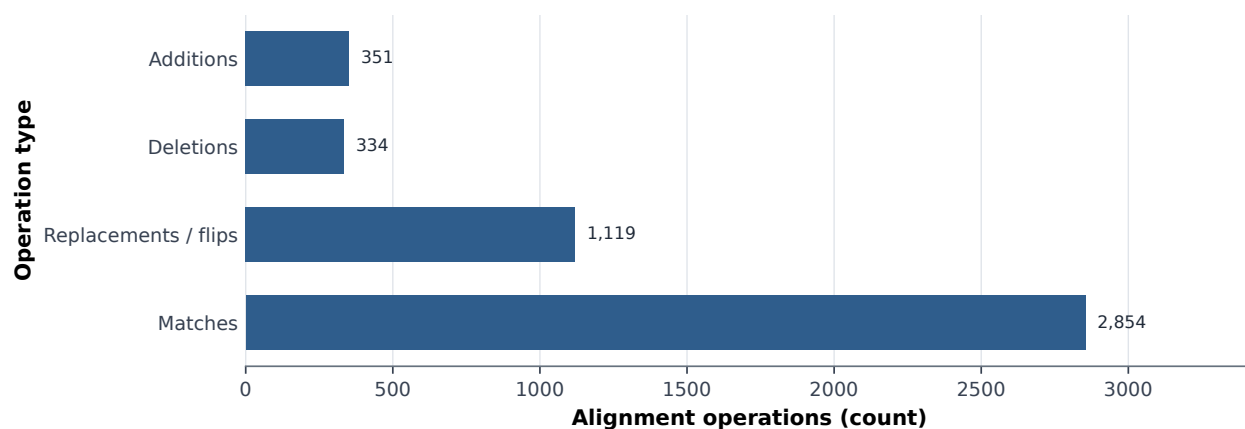


Figure 16: Horizontal bars sum each alignment operation across 595 rows: insertion repairs a missing OCR character, deletion removes an extra one, replacement changes one, and match preserves one. Replacements average 1.88 per row and matches 4.80; additions and deletions each average below 0.60.

Insights

1. Replacements are the dominant error operation, so character-confusion evidence matters more than insertion-only or deletion-only handling.
2. The average 4.80 matches per row explains why many corrupted strings still retain usable lexical evidence.

Character-alignment composition by OCR model

Each horizontal bar sums to 100% of source alignment operations for that model.

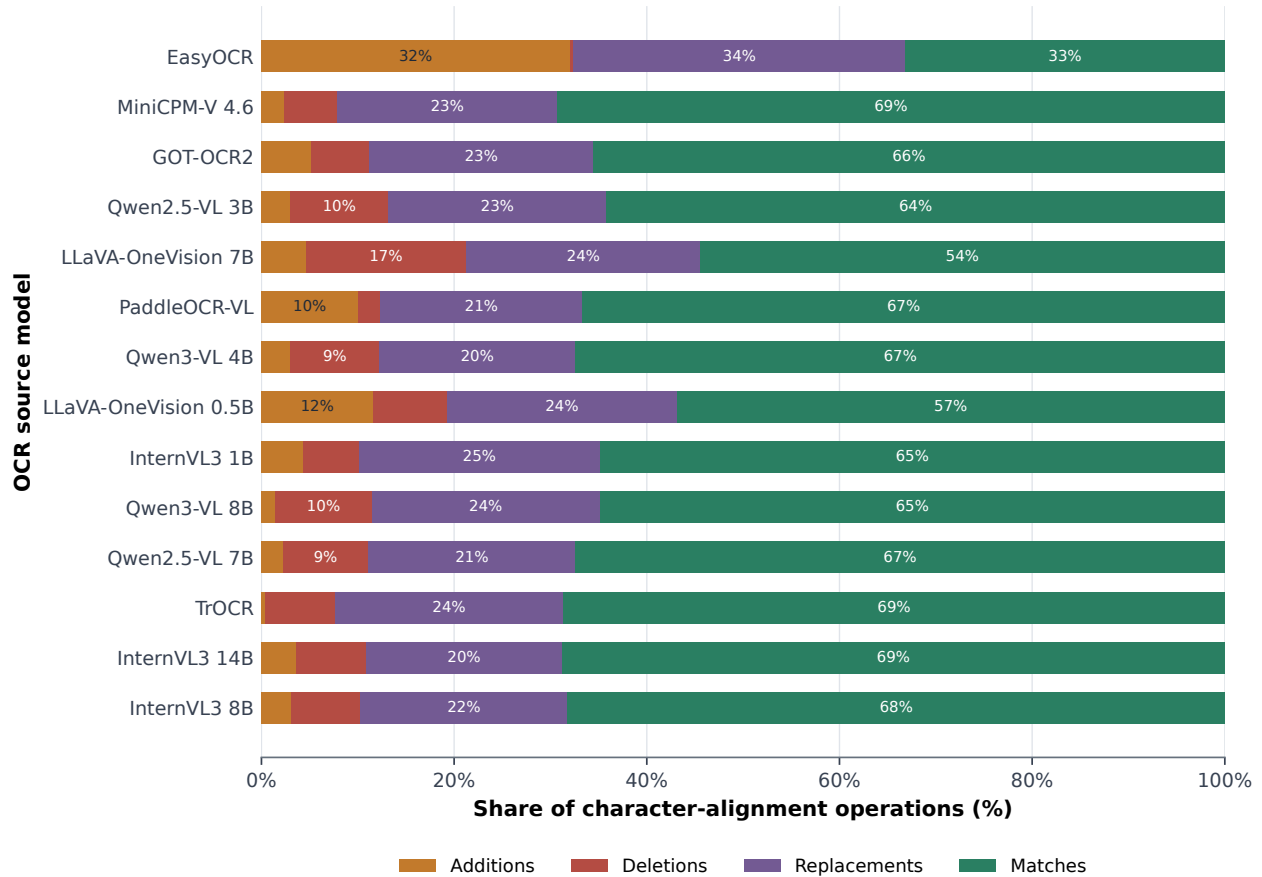


Figure 17: Each stacked bar is one OCR model and sums to 100% of its alignment operations; colors separate insertion, deletion, replacement, and match shares. EasyOCR has 33.1% matches versus 69.3% for MiniCPM-V 4.6, so raw search scores mix model behavior with different error profiles.

Insights

1. OCR models produce materially different operation mixtures, especially in their match share.
2. Any raw downstream ranking of models confounds model behavior with the type of errors assigned to each model.

Most common alignment-operation signatures

M = match, I = insertion into OCR output, D = deletion, R = replacement.

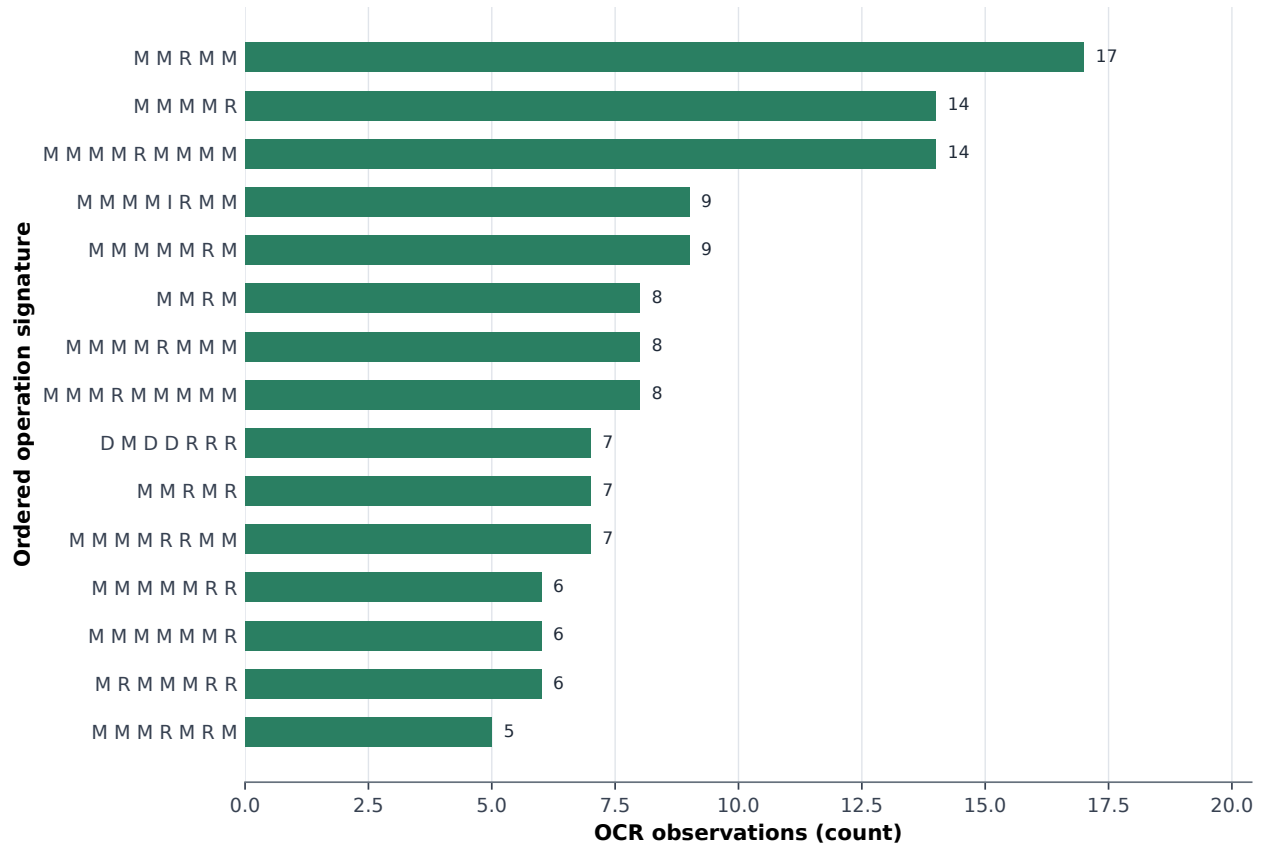


Figure 18: The y-axis lists ordered alignment signatures and the x-axis counts OCR observations; M is match, I insertion, D deletion, and R replacement. The most common signature, M M R M M, occurs 17 times, while mixed-operation sequences are dispersed across many rarer patterns.

Insights

1. No single alignment signature dominates enough rows to justify a pattern-specific correction rule.
2. The long tail of mixed signatures supports combining several retrieval signals instead of hard-coding observed strings.

The error-shape analysis shows a severe tail and many mixed-operation rows; the next section measures how often Algorithm 4 still retrieves the verified family.

6 Algorithm 4 Recovery

On the inclusive 595-row denominator, Algorithm 4 returns the verified family first for 299 rows. Another 140 place it between ranks 2 and 20. The remaining 156 omit it from the top 20. Among those 156 misses, 144 return alternative candidates and 12 return no candidates. These counts describe all rows; they are not the primary score after collision and duplicate handling.

Denominator	Cases	Hit@1	Hit@20
All observations	595	50.25%	73.78%
Scored observations, collisions removed	578	51.73%	75.61%
All unique compact query-target pairs	477	46.33%	69.60%
Scored unique pairs, primary	464	47.63%	71.12%

Table 9: There is no single accuracy without a denominator. The primary score gives one vote to each unique scored query-target pair; inclusive analyses keep all 595 observations to expose extreme and collision behavior. Cite the primary row for algorithm comparison and the inclusive row for complete-data auditing.

Cumulative Algorithm 4 recovery

Hit@k = percentage with the verified family at or before rank k; denominator is all n=595 rows.

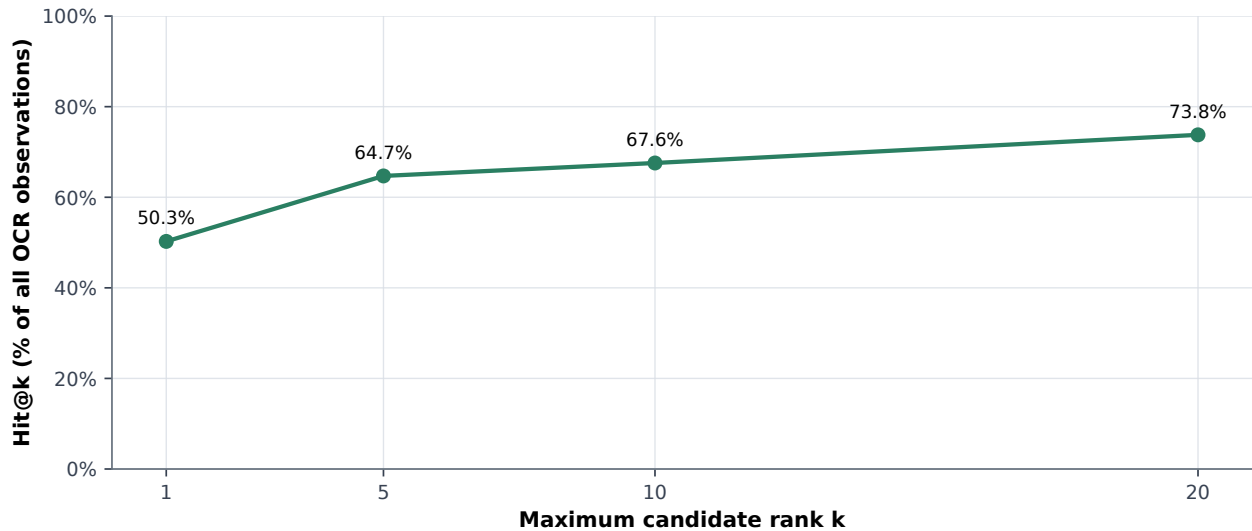


Figure 19: The x-axis increases the allowed rank cutoff and the y-axis reports inclusive recovery across all 595 rows. Recovery rises from 50.25% at rank 1 to 73.78% at rank 20; rank 5 already captures 86 of the 140 rows found below rank 1.

Insights

1. Most recoverable rank-1 misses are already near the top: 86 of 140 appear by rank 5.
2. Reranking can raise Hit@1 for those rows, but it cannot repair the 156 rows absent from the top 20.

Mutually exclusive Algorithm 4 outcomes

The three bars partition all n=595 observations.

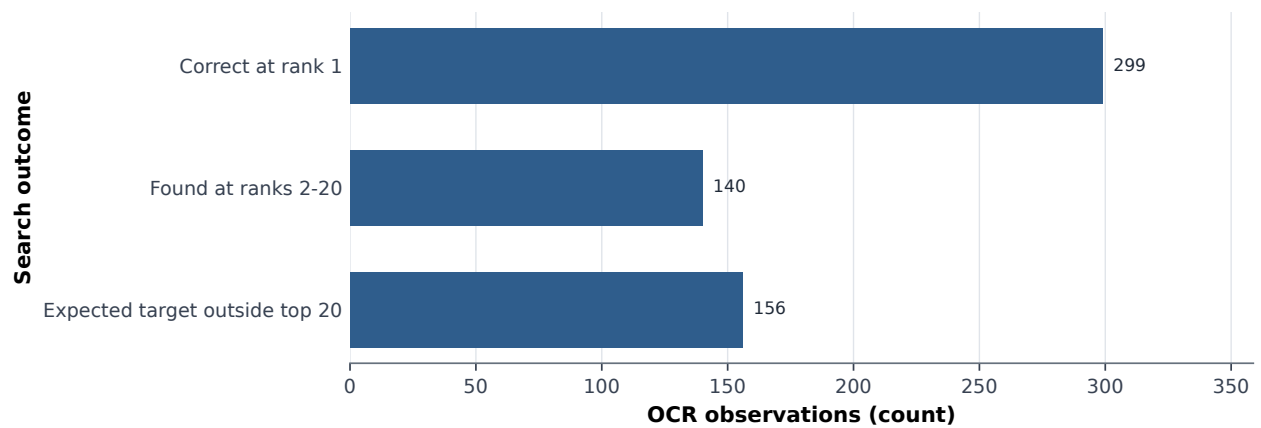


Figure 20: The three bars partition all 595 rows: 299 targets are rank 1, 140 are at ranks 2–20, and 156 are outside the top 20. Only the middle group can be fixed by reranking existing candidates; the last group needs better retrieval.

Insights

1. Ranking work targets the middle 140 rows because the verified family already exists in their candidate lists.
2. Retrieval work targets the 156 top-20 misses and has the larger implementation scope.

Recovery by analysis cohort

All 595 rows; priority: collision, exact, >0.60 extreme, fragment, >0.40 high, otherwise standard.

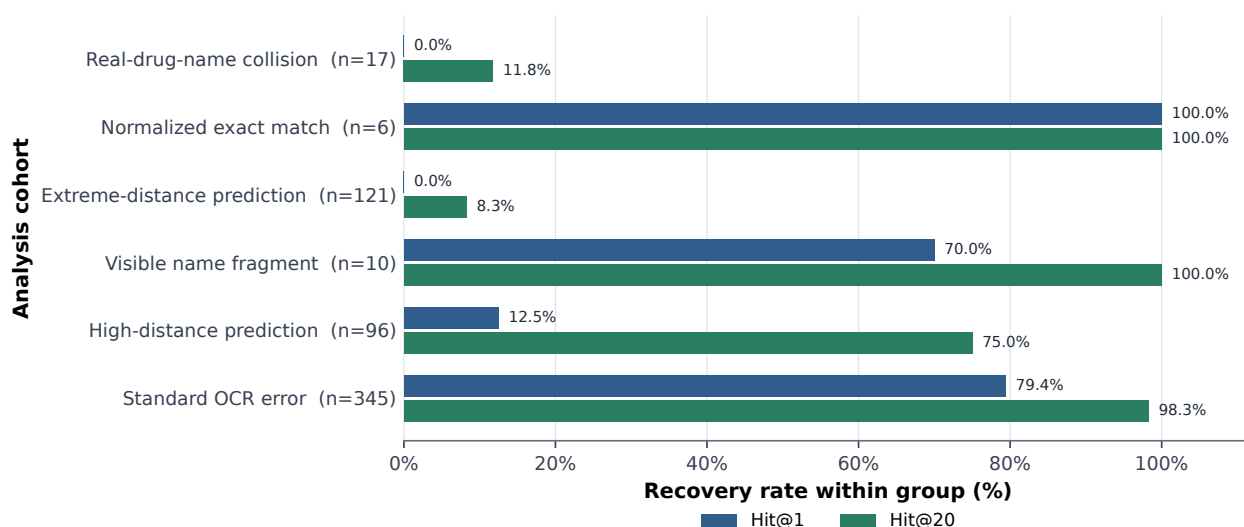


Figure 21: Blue bars are Hit@1 and green bars are Hit@20 within each cohort. Standard means $r \leq 0.40$ after earlier rules, high means $0.40 < r \leq 0.60$, extreme means $r > 0.60$, fragment means the query is a target substring, normalized exact means compact strings equal, and collision means the query exactly names another family. Standard rows reach 98.26% Hit@20, while extreme rows reach 8.26%.

Insights

1. Algorithm 4 is reliable on standard OCR errors but loses almost all recovery once corruption becomes extreme.
2. Collision performance should not drive tuning because the query itself exactly supports a different real family.

Recovery by OCR source model

Inclusive model rates; overall Hit@1=50.3% and Hit@20=73.8% across all 595 rows.

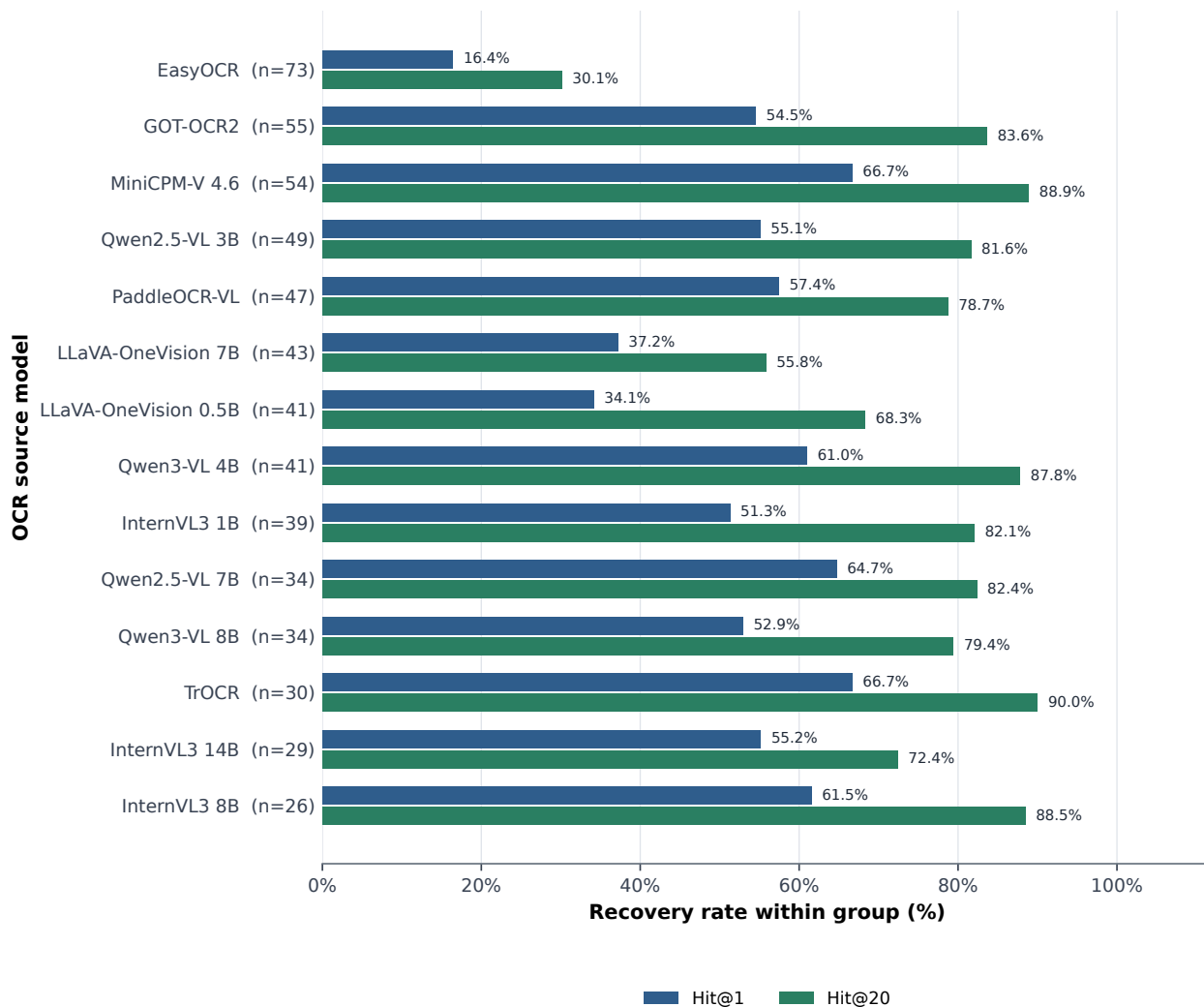


Figure 22: Blue and green bars report Hit@1 and Hit@20 within each OCR model's printed n , which ranges from 26 to 73. EasyOCR reaches 16.44% Hit@1 on its 73 rows, but 46 are extreme ($r > 0.60$); these are downstream search rates, not OCR accuracy on a shared image set.

Insights

1. Raw model rates measure downstream search on each model's supplied case mix, not OCR accuracy on shared images.
2. Models with many extreme rows need case-mix-controlled analysis before any quality claim.

Recovery by verified medicine target

Inclusive target rates; unequal n means this is not a target-balanced comparison.

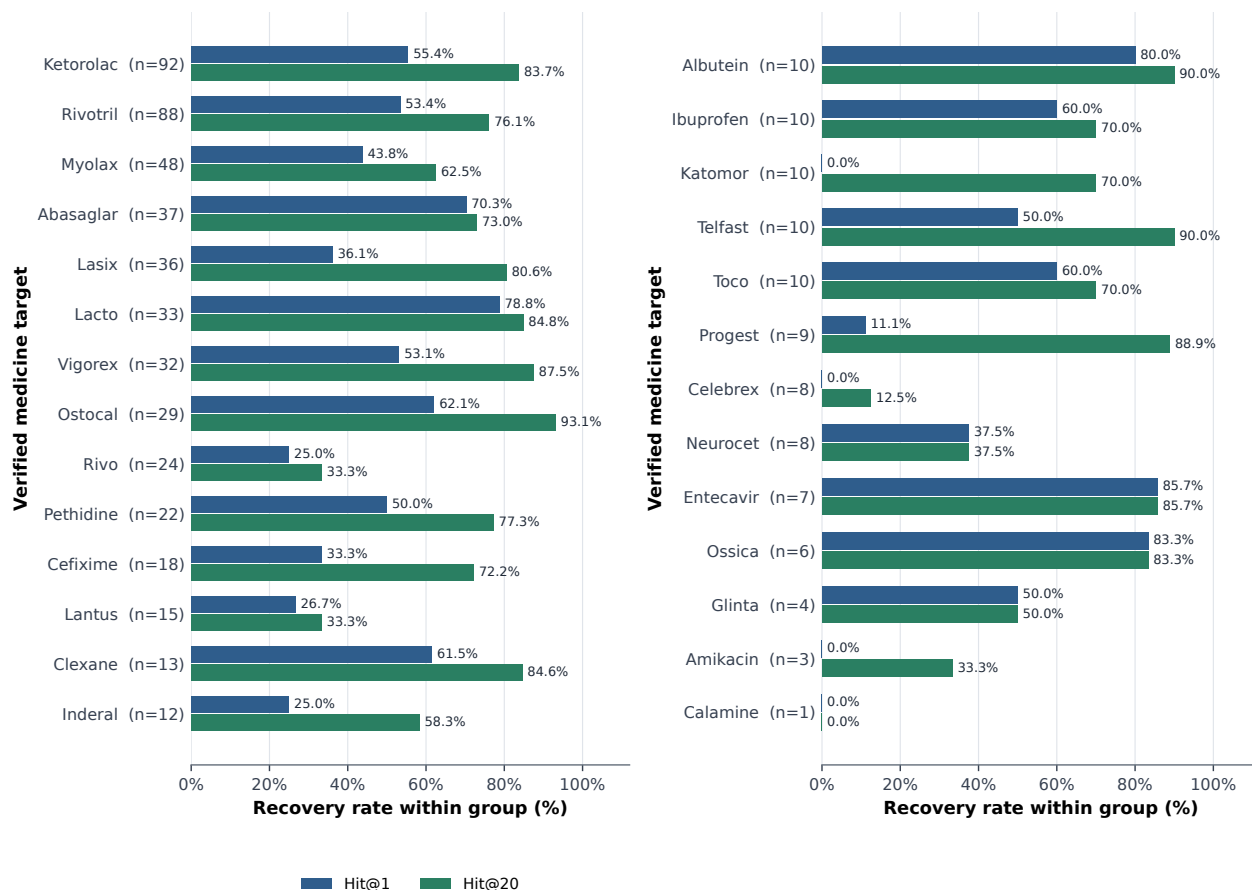


Figure 23: The inclusive denominator keeps all 595 observations, including repetitions and collisions. Blue and green bars report Hit@1 and Hit@20 within each verified target's printed n . KETOROLAC has 92 rows and 55.43% Hit@1, while CALAMINE has one row and 0%; unequal counts make this a diagnostic, not a stable target leaderboard.

Insights

1. High-count targets are the most useful starting point for reducing the benchmark's absolute miss count.
2. Low-count target percentages have high variance and should not be used to rank family difficulty.

Recovery by compact edit-distance band

Compact Levenshtein removes spaces/punctuation; bands contain 0, 1, 2-3, 4-5, or 6+ edits.

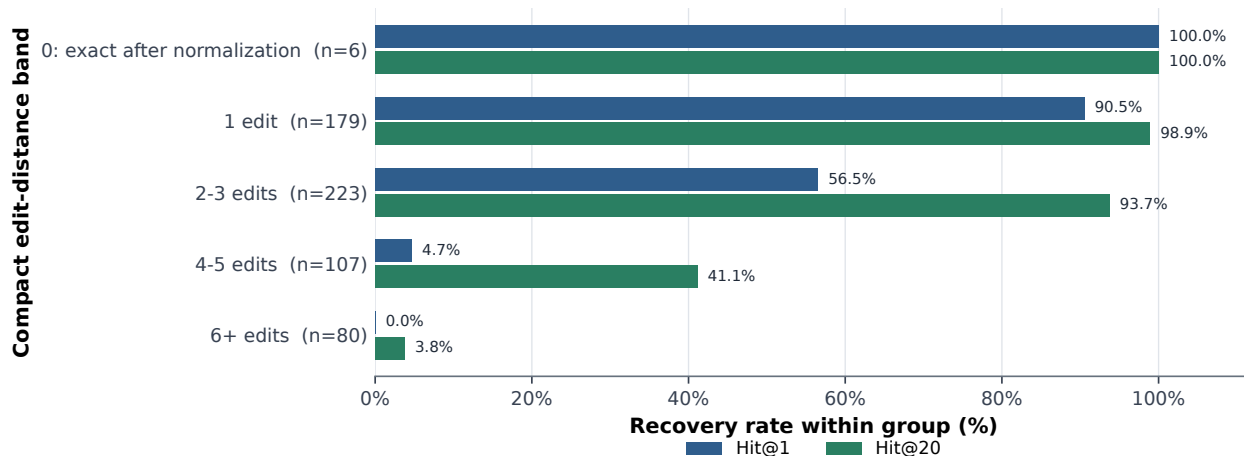


Figure 24: Blue and green bars report Hit@1 and Hit@20 within compact Levenshtein bands, where spaces and punctuation are removed before counting edits. One to three edits remain highly recoverable; the 4–5 band misses 63 of 107 rows and the 6+ band misses 77 of 80.

Insights

1. Recovery changes sharply at four compact edits, identifying a practical boundary for the current retriever.
2. Six or more edits are almost unrecoverable from the medicine string alone and need more evidence or manual review.

Recovery collapses after three compact edits; the next section isolates the 121 extreme rows to determine which models and targets create that severe tail.

7 The Extreme Cohort

The extreme cohort contains 121 rows, or 20.34% of the inclusive CSV. Each row has compact normalized distance $r > 0.60$ after collision and exact-match rules are applied. The cohort's mean r is 0.937, its median is 0.833, and its range is 0.625–2.000. Its mean compact edit count is 6.16. These predictions remain visible because they measure where lexical evidence collapses; they should not be interpreted as ordinary one- or two-character misspellings.

Extreme-distance observations by OCR model

Extreme = compact edit distance / target length > 0.60 ; all $n=121$ rows remain visible.

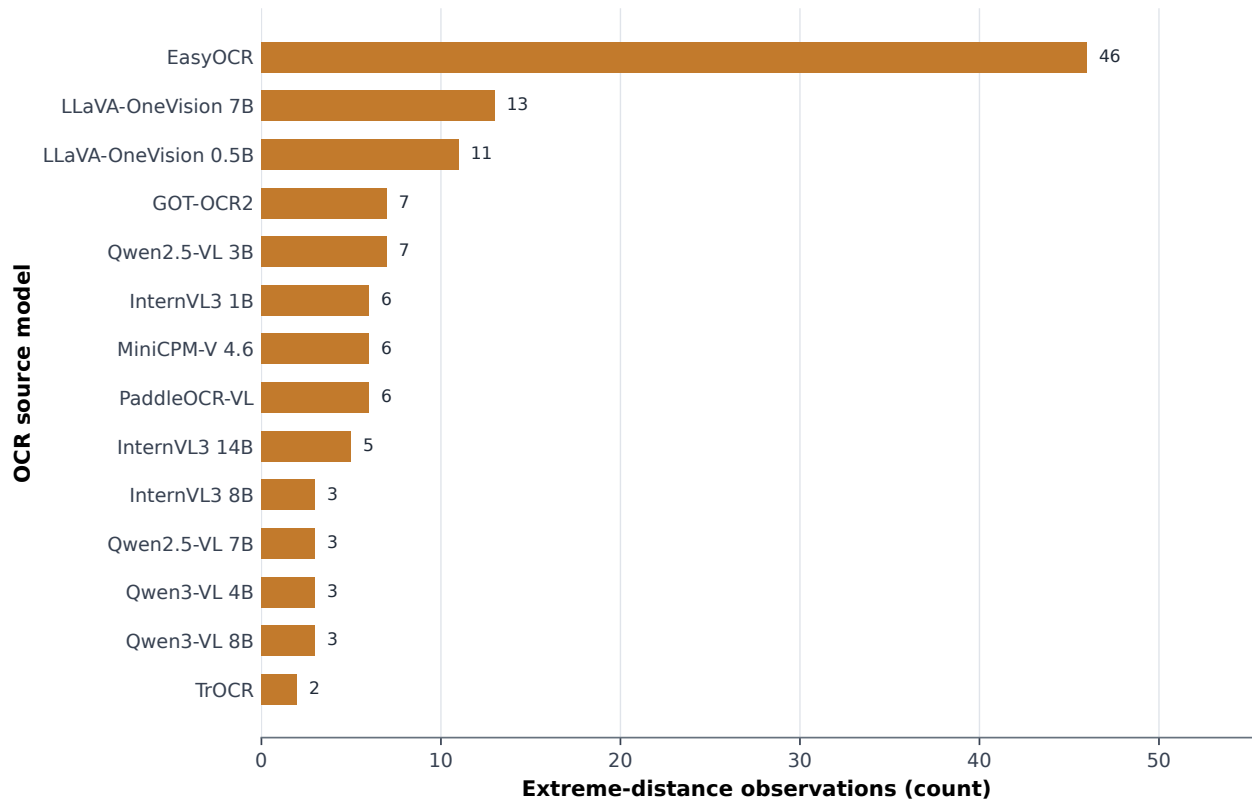


Figure 25: Bars count extreme rows by OCR model, where extreme means compact edit distance divided by target length exceeds 0.60. EasyOCR supplies 46 of 121, or 38.02%; its raw downstream score therefore reflects a more severe case mix.

Insights

1. EasyOCR supplies a disproportionate share of the severe tail, which depresses its raw downstream score.
2. Extreme-row counts identify where to audit source crops and preprocessing before comparing OCR models.

Extreme-distance observations by verified target

Extreme = compact edit distance / target length > 0.60; n=121 across displayed targets.

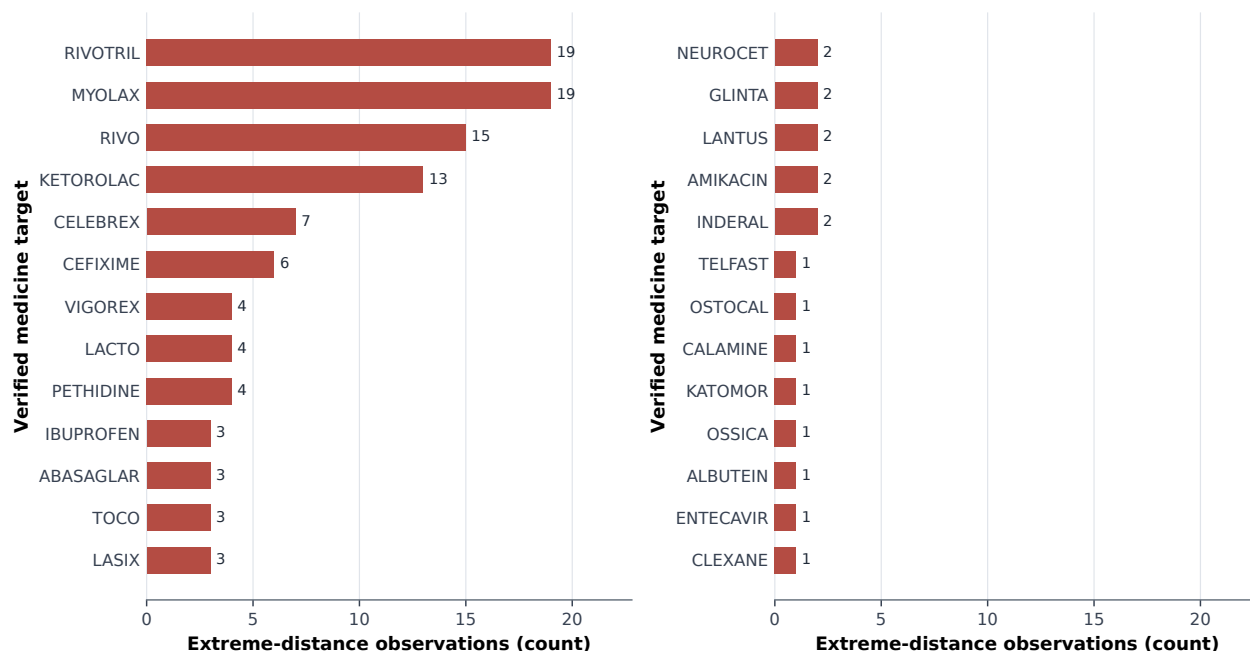


Figure 26: Bars count extreme rows ($r > 0.60$ after collision and exact-match priority) by verified target. MYOLAX, RIVOTRIL, RIVO, and KETOROLAC account for 66 of 121 rows, so auditing these four targets covers 54.55% of the cohort.

Insights

1. Auditing four target families covers 54.55% of all extreme observations.
2. Concentration by target can indicate repeated difficult images, repeated labels, or target-specific OCR confusion.

OCR input	Target	Algorithm top 1	Why review is required
ARDANCE	RIVO	JARDIANCE	Same disputed pair appears for seven OCR models.
CINNARIZIN	MYOLAX	CINNARIZINE	Output is almost an exact different catalog name.
INDOMETHACIN	INDERAL	INDOMETHACIN	Exact real drug conflicts with supplied target.
NEURONTIN	NEUROCET	NEURONTIN	Exact real drug wins over another family.

Table 10: These rows must remain in the report, but they should not drive search tuning until the source images confirm the labels.

Algorithm 4 outcomes on extreme predictions

The outcomes partition $n=121$ rows whose compact normalized distance is >0.60 .

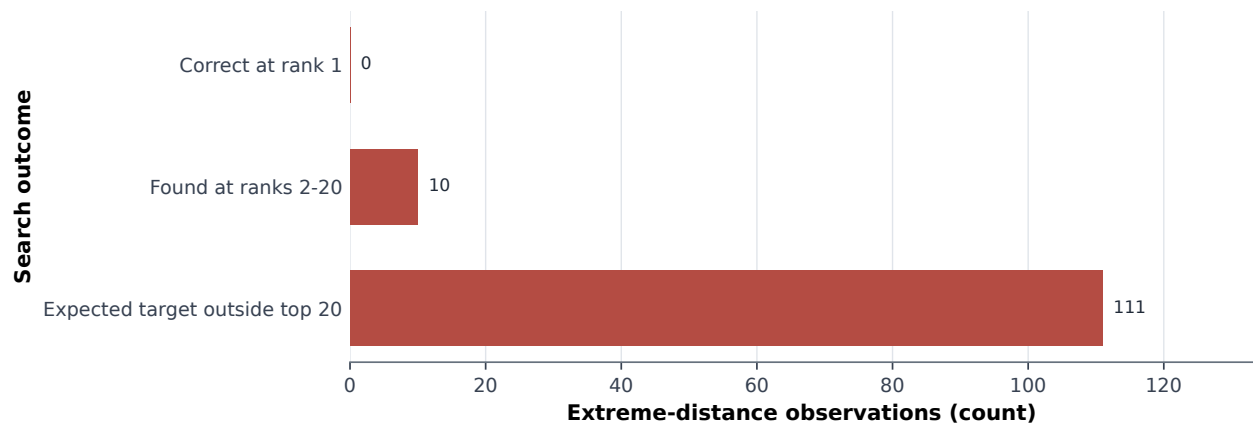


Figure 27: The three bars partition all 121 extreme rows ($r > 0.60$): zero reach rank 1, 10 reach ranks 2–20, and 111 are outside the top 20. Extreme Hit@20 is therefore 8.26%, showing that ordinary lexical retrieval rarely survives this corruption level.

Insights

1. Extreme failures are primarily retrieval failures, not small ranking mistakes.
2. Ordinary lexical matching is insufficient when more than 60% of the target length must be edited.

Compact normalized edit distance versus compact similarity

One point per row (n=595); compacting removes spaces/punctuation before Levenshtein distance.

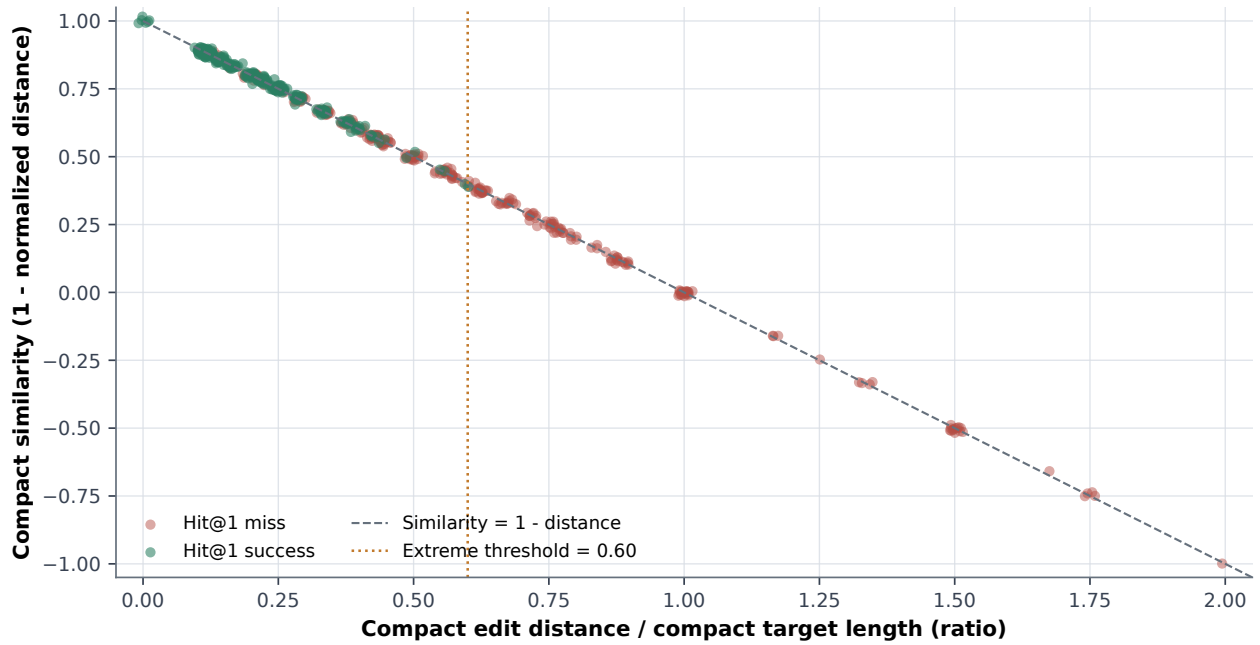


Figure 28: Each point is one row: the x-axis is compact normalized distance r , the y-axis is compact similarity $1 - r$, green marks Hit@1 success, and pink marks failure. The 0.60 vertical line is the extreme threshold; collision rows retain their collision label because exact naming of another family has priority.

Insights

1. Hit@1 success concentrates on the low-distance side of the operational 0.60 threshold.
2. Similarity and normalized distance are algebraically dependent, so the diagonal pattern is expected and is not a separate correlation finding.

The extreme audit separates low-evidence failures from disputed labels; the next section compares distance, operation mix, visible length, and case composition to locate the largest measurable losses.

8 Which Factors Cause the Most Damage

The Pearson correlation between source edit count and binary Hit@1 is -0.662 . A value near -1 means larger distance is associated with fewer successes; it does not prove causation. Mixed-operation rows create 132 of 156 inclusive top-20 misses. Queries with one to three characters have the highest within-group failure rate, while the 6–7 character group creates more absolute misses because it contains more rows.

Recovery at each source edit distance

Rates use all rows at each exact source-provided edit distance; labels show group size.

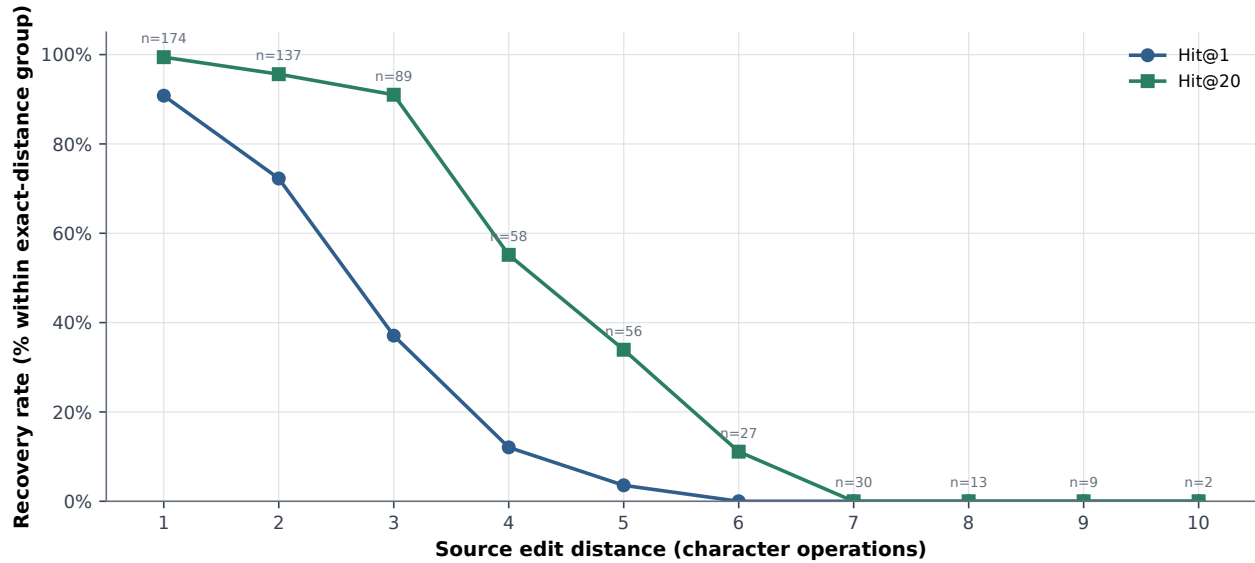


Figure 29: Blue and green bars report Hit@1 and Hit@20 at each source alignment distance, which may retain spaces. Hit@20 falls from 91.01% at distance 3 to 55.17% at distance 4 and reaches zero at distances 7–10; distance 4 is the observed recovery break.

Insights

1. The 35.84-point Hit@20 drop from distance 3 to 4 is the clearest observed recovery break.
2. Zero recovery at distances 7–10 indicates missing evidence rather than a near-tie ranking problem.

Recovery by character-operation profile

Mixed = at least two of insertion, deletion, and replacement counts are nonzero in one row.

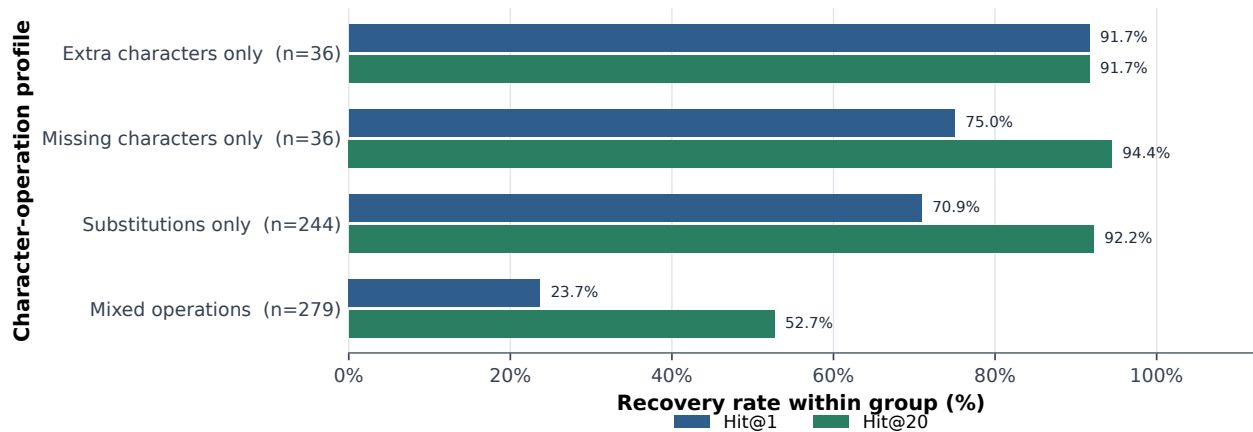


Figure 30: Blue and green bars report Hit@1 and Hit@20 within operation profiles. Mixed means at least two of insertion, deletion, and replacement occur in one row; its 279 rows reach 52.69% Hit@20 versus at least 91.67% for each isolated operation profile.

Insights

1. Mixed-operation rows combine several corruption mechanisms and are far harder than any isolated operation profile.
2. Operation-aware candidate generation is a higher-value target than separate rules for insertion-only or deletion-only cases.

Recovery by OCR-output length

Length is source edited_length, including source spaces; fixed bands are shown on the y-axis.

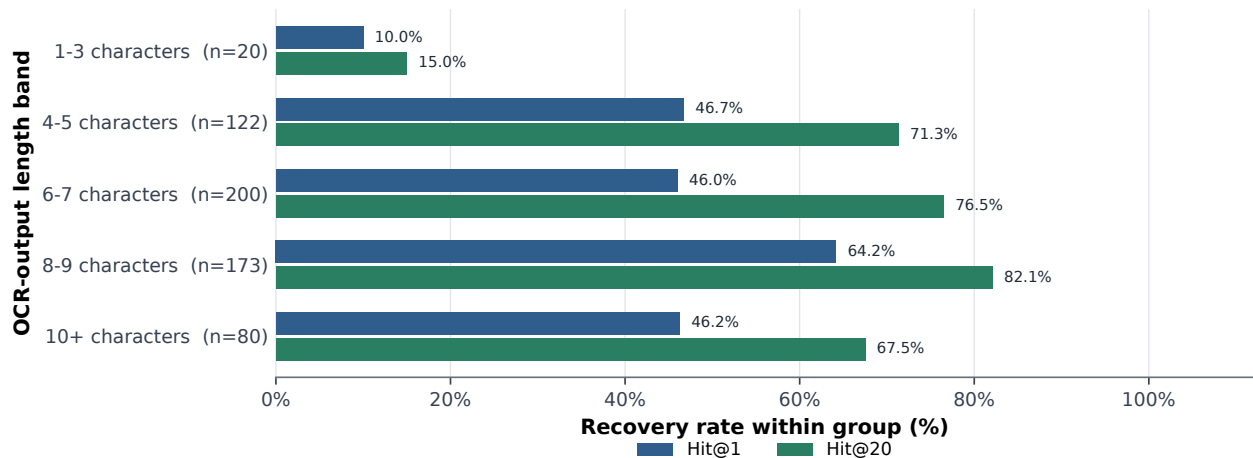


Figure 31: Blue and green bars report recovery within source OCR-length bands, including retained spaces. The 1–3 band reaches 10.00% Hit@1 and 15.00% Hit@20, while 8–9 characters reach 64.16% and 82.08%; visible length is useful evidence but does not guarantee correctness.

Insights

1. Very short outputs should trigger a request for more visible characters because their top-20 recovery is only 15%.
2. The 8–9 character band performs best, showing the value of retained evidence even when some characters are wrong.

Recovery by output-versus-target length direction

Difference = target length - OCR length: positive means OCR shorter, zero means equal.

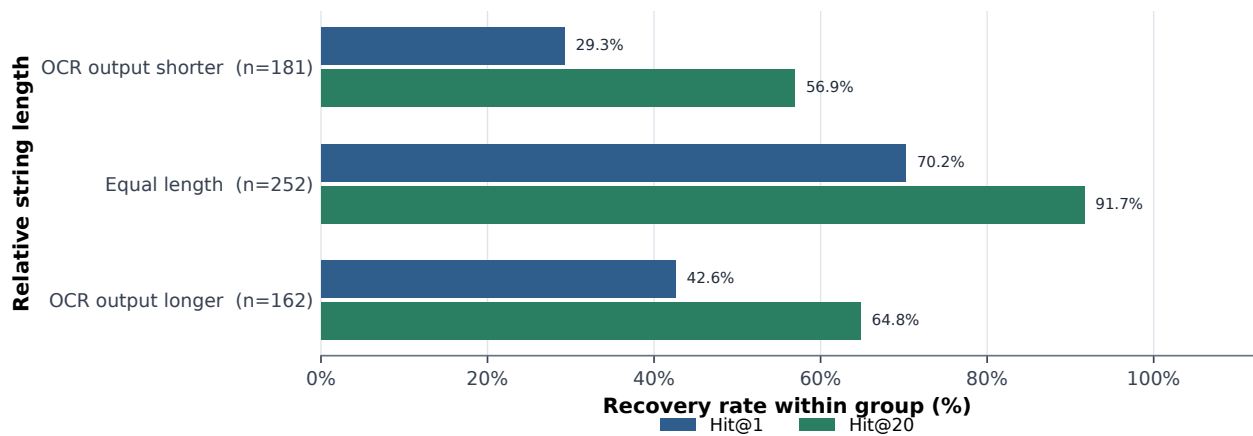


Figure 32: Groups use source target length minus OCR length: positive means OCR is shorter, zero means equal, and negative means longer. Equal-length outputs reach 91.67% Hit@20, versus 56.91% when shorter and 64.81% when longer, although equal length can still hide substitutions.

Insights

1. Insertions and deletions that change length damage recovery more than equal-length corruption on this dataset.
2. Equal length is useful supporting evidence but cannot distinguish a correct string from replacement-heavy corruption.

Raw versus standard-error-only model recovery

Standard errors have normalized distance ≤ 0.40 and are not exact, fragments, or collisions.

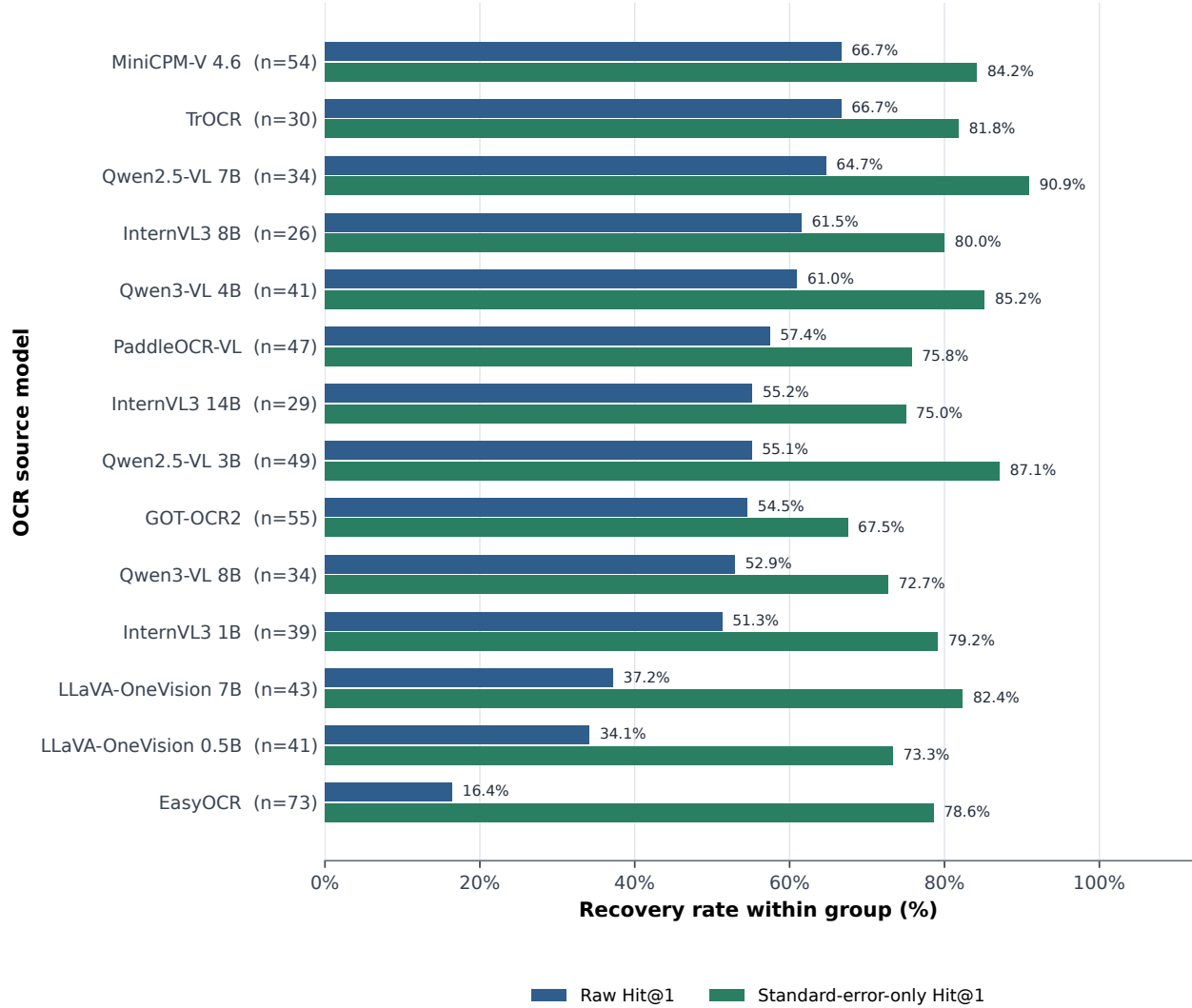


Figure 33: For each OCR model, blue uses all supplied rows and green keeps only standard rows, where r is compact edit distance divided by target length and $r \leq 0.40$ after exact, fragment, and collision rules. EasyOCR rises from 16.44% raw Hit@1 to 78.57% standard-only Hit@1 because 46 of its 73 rows are extreme; raw model ordering therefore confounds quality and case mix.

Insights

1. Model ordering changes substantially after controlling for standard-error severity.
2. A fair OCR comparison requires the same images, targets, crops, and preprocessing for every model.

Metric changes under four denominators

Scored excludes 17 real-drug collisions; unique counts each compact query-target pair once.

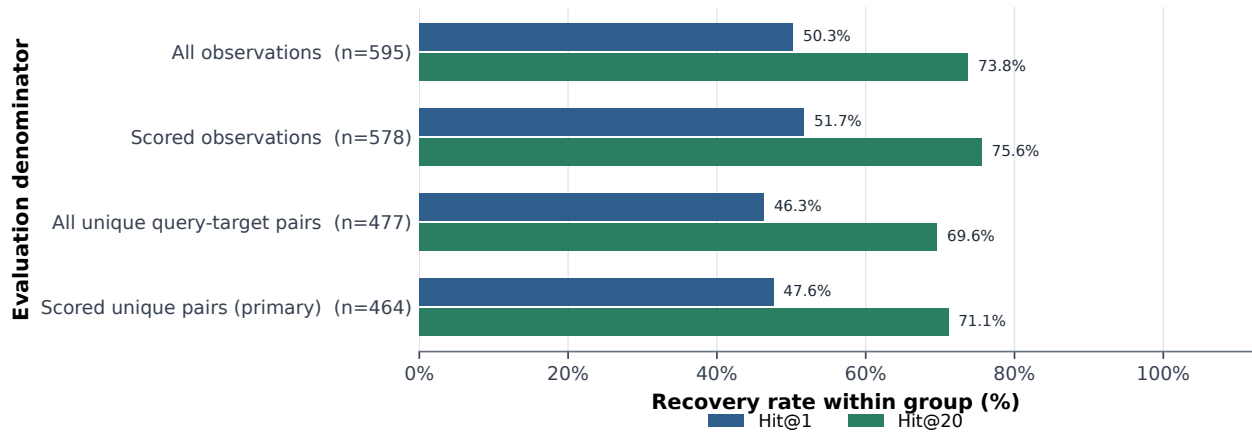


Figure 34: Each row is one valid denominator; blue is Hit@1 and green is Hit@20. “Scored” removes 17 real-drug collisions and “unique” gives each compact query-target pair one vote. Cite the primary scored-unique result: 47.63% Hit@1 and 71.12% Hit@20 over 464 pairs.

Insights

1. Duplicate observations and collision policy change the measured rate by changing row weights and eligibility.
2. The primary denominator is appropriate for algorithm comparison; the inclusive denominator is appropriate for data auditing.

Primary recovery by development and holdout split

Primary denominator: 464 scored unique pairs; target hashing keeps one family in one split.

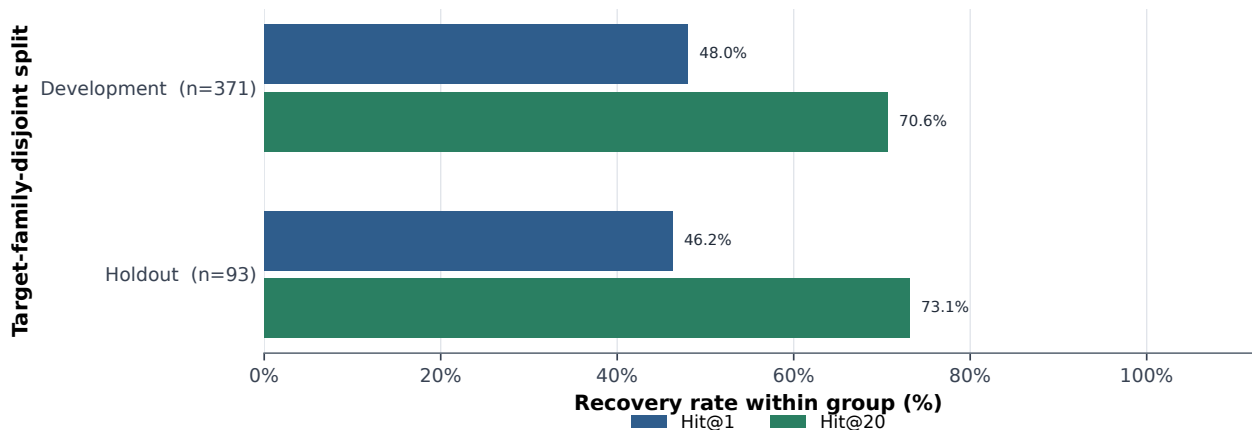


Figure 35: Blue and green bars report primary Hit@1 and Hit@20 after target-family hashing keeps each family in one split. Development has 371 unique pairs, including KETOROLAC; holdout has 93, including VIGOREX. Holdout reaches 46.24% Hit@1 and 73.12% Hit@20, so its larger loss is ranking rather than retrieval.

Insights

1. Holdout Hit@20 remains close to the overall primary result, supporting retrieval generalization to unseen families.
2. The larger Holdout Hit@1 gap points to ranking quality rather than failure to generate the verified family.

These factor comparisons identify weak groups but not their total cost; the next section separates within-group failure rate from each group’s share of all misses.

9 Failure Impact

Failure rate answers which group is weakest. Failure share answers where the largest number of misses comes from. If group g contains n_g rows and F_g top-20 misses, the orange bar is F_g/n_g and the red bar is $F_g/156$. For the final rank-only figure, the red denominator changes from 156 retrieval misses to 140 rank-only failures. Engineering priority should use both rate and absolute impact.

Top-20 failure rate and total impact by cohort

Orange divides misses by cohort rows; red divides cohort misses by all top-20 misses.

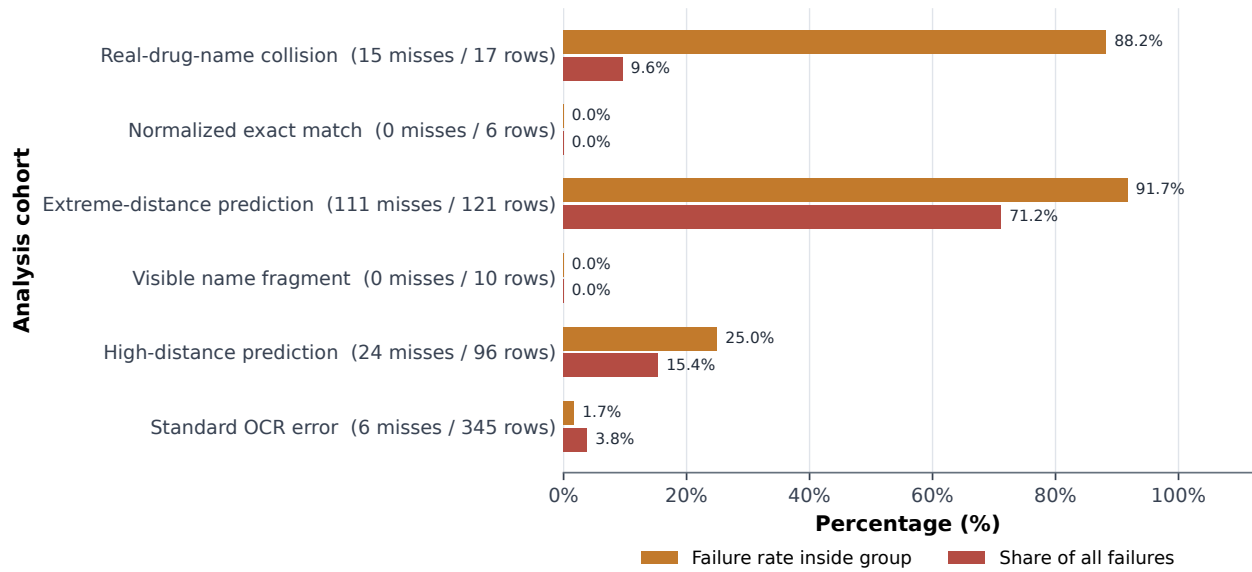


Figure 36: Cohorts apply collision and compact-exact priority, then extreme ($r > 0.60$), fragment, high ($0.40 < r \leq 0.60$), and standard ($r \leq 0.40$). Orange is misses divided by cohort rows and red is misses divided by all 156 inclusive failures. Extreme misses 111 of 121 rows, giving the highest rate, 91.74%, and largest impact, 71.15%.

Insights

1. Extreme corruption is both the weakest cohort and the largest source of absolute top-20 misses.
2. Source-label and crop audits should precede algorithm tuning because several extreme rows contain disputed evidence.

Top-20 failure rate and total impact by compact distance

Rate measures subgroup difficulty; share measures contribution to the benchmark's total failures.

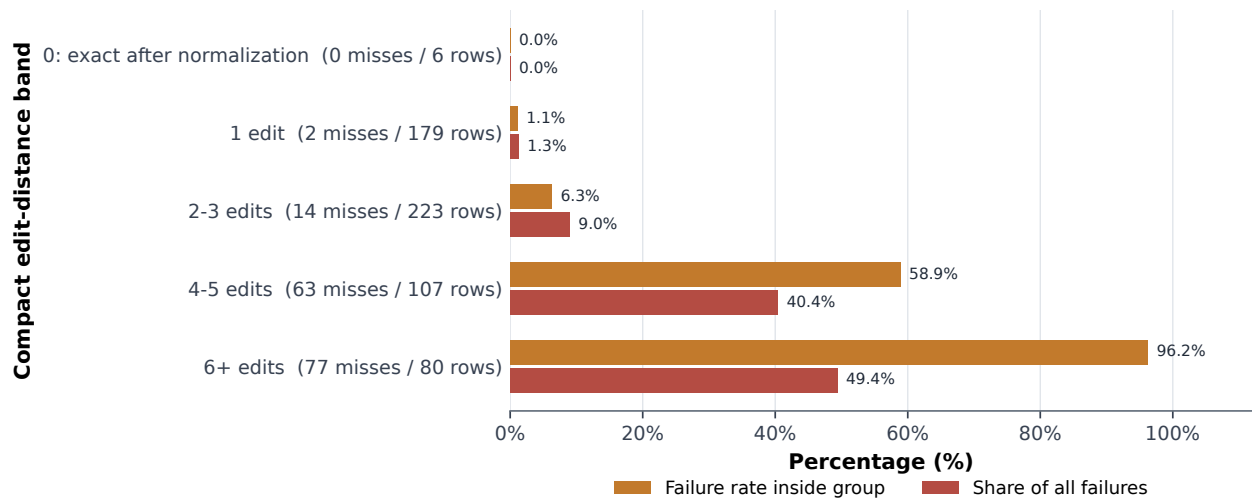


Figure 37: Orange is within-band failure rate and red is share of all 156 misses; compact distance removes spaces and punctuation before counting edits. Distance four or more creates 140 misses, or 89.74%; the 6+ band misses 77 of 80 rows.

Insights

1. Compact distance four or more accounts for 89.74% of all retrieval failures.
2. Improving ordinary one-edit handling alone cannot materially change the benchmark's dominant failure count.

Top-20 failure rate and total impact by operation profile

Rate and impact use the same 156 top-20 failures but answer different questions.

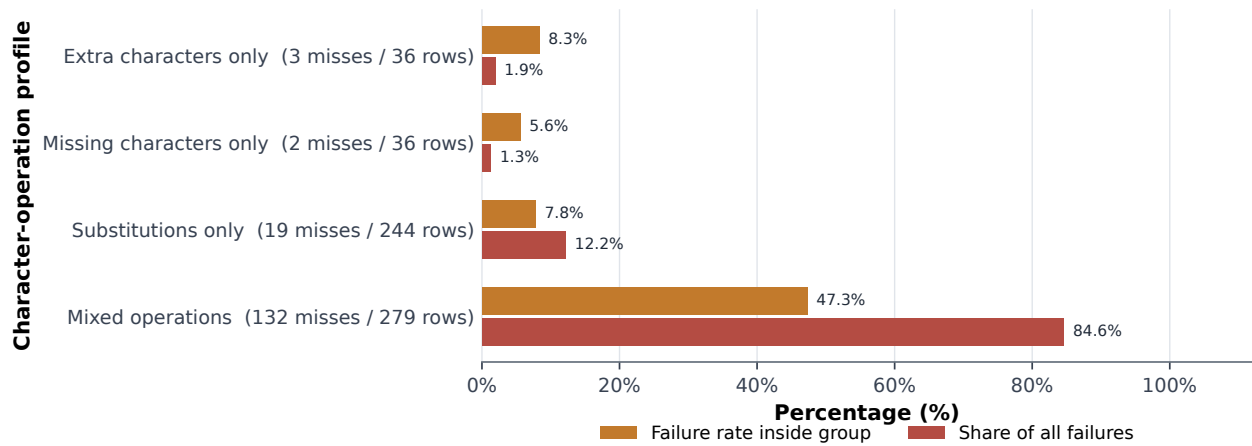


Figure 38: Orange is within-profile failure rate and red is share of all 156 misses. Mixed means at least two of insertion, deletion, and replacement occur; these rows create 132 misses, or 84.62%, while isolated profiles create 24.

Insights

1. Mixed operations create 84.62% of all top-20 misses and therefore offer the largest operation-level improvement opportunity.
2. The retriever should combine insertion, deletion, and replacement evidence within one query rather than process them as isolated cases.

Top-20 failure rate and total impact by OCR-output length

Short strings can have high failure rates without producing the largest absolute miss count.

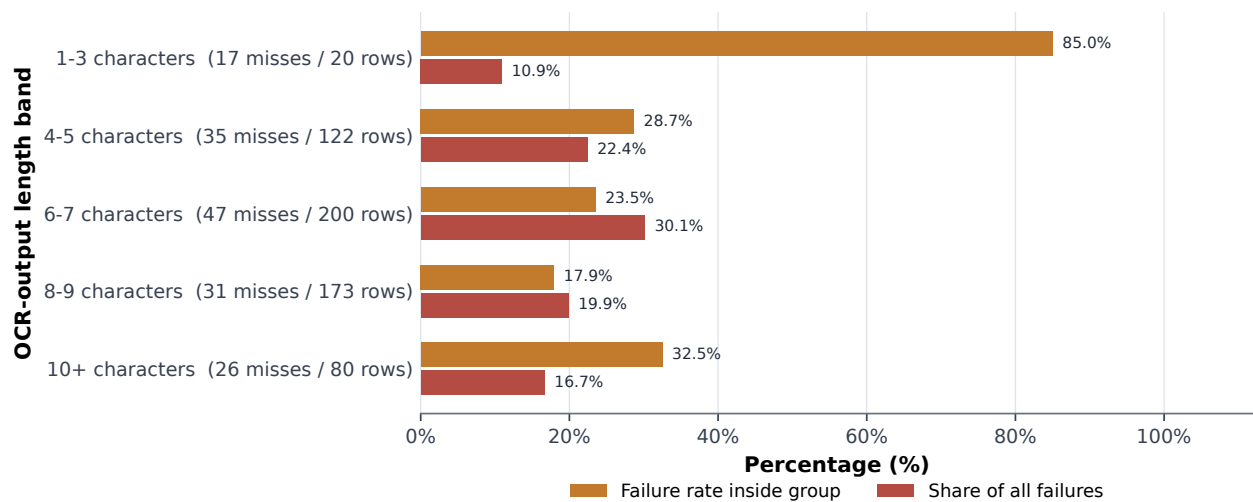


Figure 39: Orange is failure rate within each source OCR-length band, whose length retains source spacing; red is share of all 156 misses. The 1–3 band misses 17 of 20 rows, the worst rate at 85.00%; the 6–7 band creates the most misses, 47, because it has 200 rows.

Insights

1. The 1–3 character group is the weakest by rate, so it needs a safer user interaction that requests more evidence.
2. The 6–7 character group creates the most misses by count, so algorithm work there has the larger aggregate payoff.

Top-20 failure rate and total impact by OCR model

Model-level impact combines model quality, observation count, target mix, and error severity.

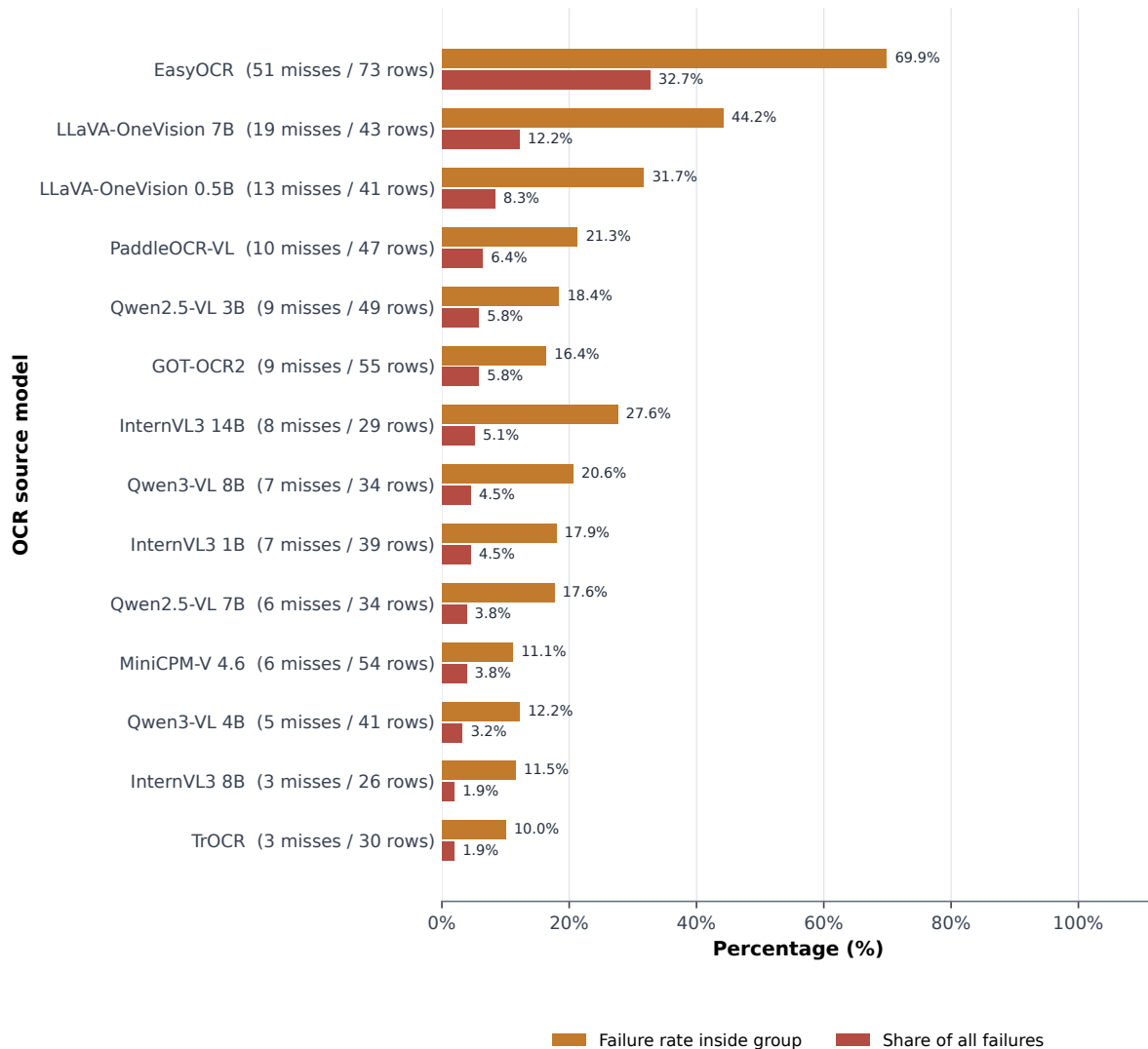


Figure 40: Orange is top-20 failure rate within each model and red is that model's share of all 156 misses. EasyOCR misses 51 of 73 rows and contributes 46 extreme rows ($r > 0.60$), so its 32.69% failure share is not an image-matched OCR quality comparison.

Insights

1. EasyOCR is the largest observed source of misses, but its extreme-heavy case mix prevents attributing that impact to model quality alone.
2. Model-level engineering decisions require a matched-image rerun rather than these unequal observational groups.

Targets with the largest top-20 failure impact

The figure shows the 12 targets contributing the most misses; red bars sum against all 156 misses.

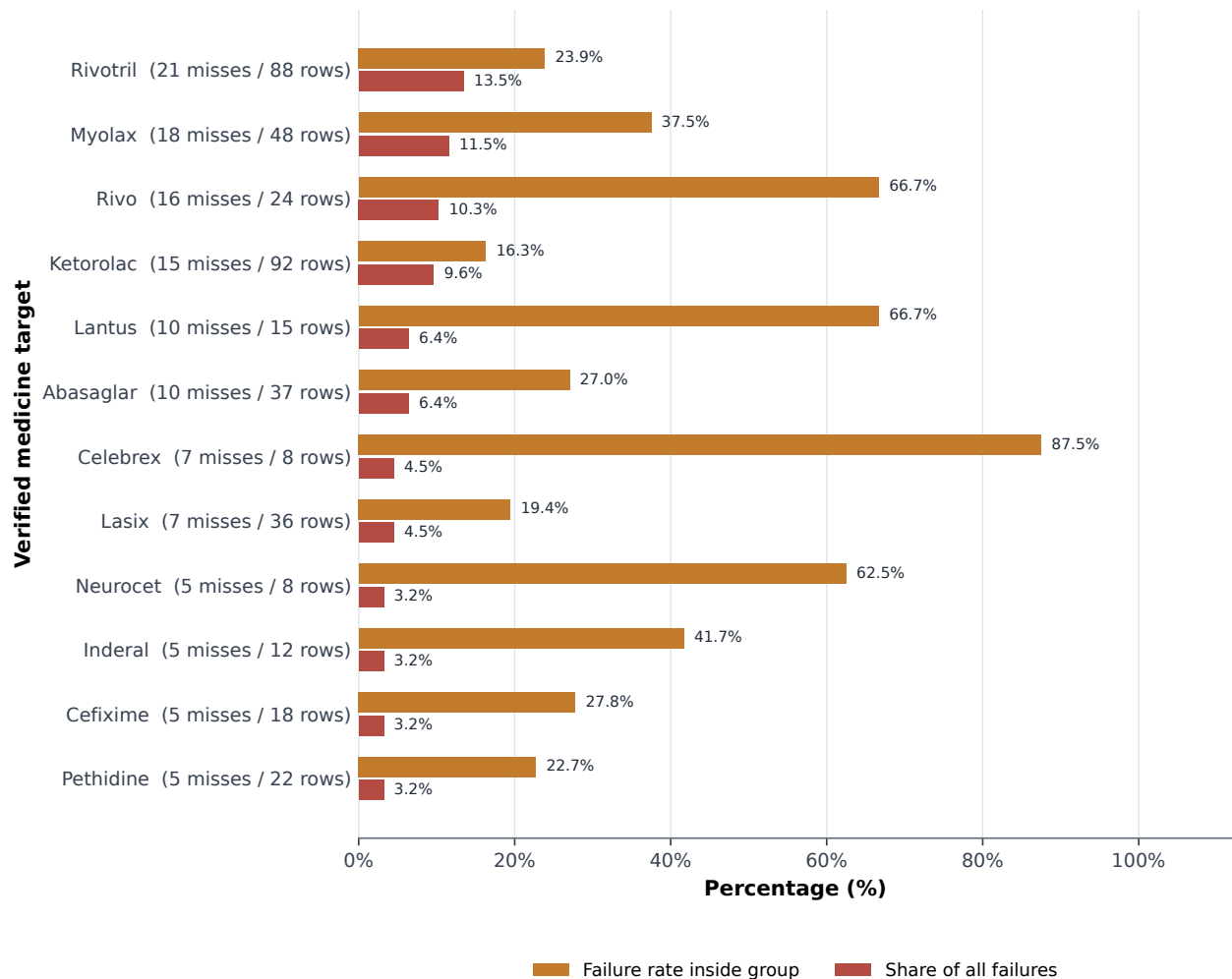


Figure 41: Orange is failure rate within each verified target and red is its share of all 156 misses; only the 12 largest contributors are shown. RIVOTRIL, MYOLAX, RIVO, and KETOROLAC create 70 misses, or 44.87%, but their rates differ because their observation counts differ.

Insights

1. Four high-volume targets account for 44.87% of all top-20 misses and provide an efficient audit starting point.
2. Failure share must be read with target exposure because frequently observed families naturally contribute more misses.

Rank-only failure rate and impact by cohort

A rank-only failure means the target exists at ranks 2-20 but not rank 1; impact uses all rank-only failures.

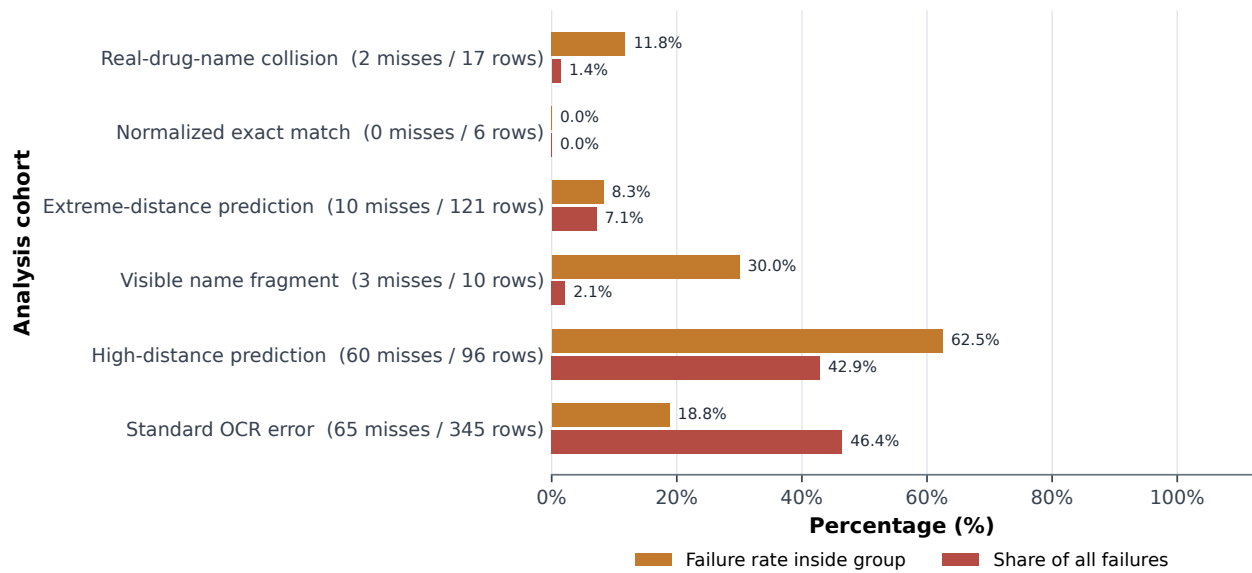


Figure 42: A rank-only failure has the verified family at ranks 2–20 but not rank 1. Cohorts apply collision and compact-exact priority, then extreme ($r > 0.60$), fragment, high ($0.40 < r \leq 0.60$), and standard ($r \leq 0.40$). Orange divides failures by cohort rows and red by all 140 rank-only failures; standard creates 65 and high-distance creates 60, so reranking can address both groups.

Insights

1. Standard and high-distance cohorts create 125 of 140 rank-only failures, so reranking benefits both mild and moderate corruption.
2. These cases are lower-risk implementation targets because the verified family already exists inside the top 20.

Failure rate and failure share now identify both difficult and high-impact groups; the next section converts that evidence into ordered engineering work.

10 Engineering Decisions

Across all 595 rows, Algorithm 4 returns 583 *ambiguous* responses, meaning it shows candidates but does not certify one as safe. The other 12 are *no-match* responses with zero returned candidates. An unsafe confident top-1 would certify a wrong first family without requesting clarification; the count is zero. Safety therefore comes from candidate comparison rather than automatic certainty.

Order	Action	Evidence	Success condition
P0	Audit repeated extreme pairs against source images.	Several outputs are exact real names that conflict with supplied labels.	Confirm labels before algorithm tuning.
P1	Tune ranks 2–5.	86 of 140 rank-only failures resolve by rank 5.	Raise H@1 without lowering H@20 or safety.
P2	Add operation-aware retrieval for mixed errors.	Mixed operations create 132 retrieval misses.	Raise mixed-error H@20 on locked holdout data.
P3	Ask for more visible evidence on short strings and collisions.	One to three characters reach only 15.00% H@20.	Avoid forced family selection.
P4	Compare OCR models on the same image set.	Raw model results correlate with severe-error case mix.	Use matched images, targets, and preprocessing.

Table 11: The first action protects label integrity. The next two target the largest fixable ranking and retrieval losses without hard-coding medicine names.

This benchmark does not measure full-prescription OCR, dosage extraction, clinical correctness, or whether every supplied label is valid. Those require original image IDs, crop lineage, matched-model inference, and human label audit.

The conclusions depend on reproducible source rows, search outputs, and figure generation; the final artifact list identifies those exact files.

11 Artifacts

- Source observations: `benchmark_03_ocr/data/04_model_predictions/predictions.csv`
- Row-level search results: `benchmark_03_ocr/artifacts/04_model_predictions/algorithm_4_results.csv`
- Figure generator: `benchmark_03_ocr/build_model_prediction_paper_figures.py`
- Vector figures used here: `benchmark_03_ocr/results/04_model_predictions/figures/`